



Documaker

Documaker Server Installation Guide

version 11.3

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Chapter 1

Introduction

This manual provides instructions for installing the Documaker Server software on computers running OS/390 (MVS), UNIX/Linux, and 32-bit Windows, such as Windows 2000 and Windows XP.

This table will help you find the information you need:

To install	See
Documaker Server on Windows	Installing Documaker Server on Windows on page 95
Documaker Server on OS/390 (MVS)	Installing Documaker Server on OS/390 on page 1
Documaker Server on UNIX/Linux	Installing Documaker Server on UNIX/Linux on page 79

A summary of the system requirements follows.

SYSTEM REQUIREMENTS

Skywire Software applications run on a variety of operating systems and hardware platforms. Make sure you have these components before you install the following Skywire Software applications.

Operating systems

The following applications run on a variety of operating systems, principally Windows 32-bit operating systems such as Windows 2000, Windows 2003, Windows Vista, and Windows XP, UNIX/Linux 32-bit operating systems such as AIX, Solaris, and Linux x86, and z/OS (OS/390). This table shows the various product offerings and the operating systems under which they run.

NOTE To store a Documaker version 11.0 resource library in Documanager, you must have Documanager version 6.3 SR 2 or version 6.4 SR 1 or higher.

	Windows+ z/OS (OS/390)		AIX	Linux	Solaris
Docucreate	Yes	No	No	No	No
Documaker Studio	Yes	No	No	No	No
Documaker Workstation	Yes	No	No	No	No
Docupresentment	Yes	No	Yes	Yes	Yes
PDF 417++	Yes	Yes	Yes	Yes	Yes
iPPS (COM+)++++	Yes	No	No	No	No
iDocumaker Workstation (Java)+++	Yes	No	Yes	Yes	Yes
Documaker Server					
GenTrn	Yes	Yes	Yes	Yes	Yes
GenData	Yes	Yes	Yes	Yes	Yes
GenPrint	Yes	Yes	Yes	Yes	Yes
GenWIP	Yes	Yes	Yes	Yes	Yes
GenArc	Yes	Yes	Yes	Yes	Yes

+ Includes Windows 2000, Windows 2003, Windows XP, Windows Vista, Windows 2000 Server, and Windows 2003 Server.

++ Beginning with version 11.2, PDF417 is included as a component of Documaker Server.

+++ Runs under any operating system that supports the Java Virtual Machine.

++++ Runs under Microsoft Windows 2000 Server and Windows 2003 Server.

		Windows+ z/OS (OS/390)		AIX	Linux S0laris	
Printers**						
	AFP	Yes	Yes	Yes	Yes	Yes
	GDI	Yes	No	No	No	No
	HTML	Yes	No	Yes	Yes	Yes
	Metacode	Yes	Yes	Yes	Yes	Yes
	PCL	Yes	No	Yes	Yes	Yes
	PCL 6*	Yes	No	Yes	Yes	Yes
	PDF	Yes	Yes	Yes	Yes	Yes
	PostScript	Yes	Yes	Yes	Yes	Yes
	RTF	Yes	No	Yes	Yes	Yes
	VIPP	Yes	Yes	Yes	Yes	Yes
	XML**	Yes	Yes	Yes	Yes	Yes

+ Includes Windows 2000, Windows 2003, Windows XP, Windows Vista, Windows 2000 Server, and Windows 2003 Server.

* You must have PCL 6 or higher for Unicode support on PCL-compatible printers. PCL 6 support became available in version 10.2.

** Printer support depends on licensing. For example, PDF and HTML are licensed separately for the PPS market and PDF is licensed separately for the z/OS market.

Networks

The system does not use any specific network calls and is expected to work on any network compatible with Microsoft programs. At Skywire Software we use Windows servers and, to a lesser extent, Novell servers.

NOTE The network file server you use with Documaker Studio or Documaker Workstation must be a 100% Windows network compatible. Some UNIX systems that offer NFS support are not 100% Windows compatible and some UNIX systems do not honor Windows file locking calls and may not be suitable for use as a file server in a true multi-user environment.

Hardware This table outlines the minimum hardware Skywire Software uses to test the system on a single user Windows workstation. Specific requirements for running the rules processing components of Documaker on other operating systems follow. We suggest you run the system on a similar or better computer.

	Minimum Test Hardware
CPU	Pentium II 400 MHz
Memory	512mb RAM
Hard disk*	250mb
Printer	HP-compatible printers supporting PCL5 or higher
Printer memory	8mb RAM
Monitor	Color SVGA, 17-inch screen monitor

* Additional space may be required for your customized forms.

DOCUMAKER SERVER REQUIREMENTS

Your computer must have certain software and hardware components to run the programs that comprise the Documaker Server system. Depending on your software license, operating environment, and the market your solution was created for, these requirements vary.

The following tables outline the minimum hardware Skywire Software uses to test Documaker Server. We suggest that you run the system on a similar or better configuration.

NOTE For more specific information on the GenArc program and the additional archive and retrieval capabilities available from Skywire Software, refer to the [Documaker Server System Reference](#)

Should your company have special needs, contact your sales representative and keep in mind that, by using upload and download programs, additional functionality is available.

On the z/OS (OS/390) Platform

Skywire Software's DAP and Documaker Server products run on the following versions/releases of IBM's operating systems:

- OS/390 version 1.1 to version 2.10
- z/OS version 1.6 and higher

NOTE Following OS/390 version 2.10, new versions were named z/OS. Documaker runs on OS/390 and z/OS. In this manual, OS/390 and z/OS are referred to as z/OS unless otherwise noted.

No product upgrades are required and no incompatibility problems have been reported when running Documaker Server on any of these operating system releases.

	Requirements
Hard disk	150mb
Printer	Any printer which supports IBM AFP, Xerox Metacode, or Adobe PostScript
Runtime library	IBM Language Environment for OS/390 version 2.10 or higher
Compiler	(Only necessary if adding custom code to the system) IBM C/C++ Compiler for OS/390 version 2.10 or higher

NOTE Regardless of the type of computer you run the system on, to print charts on Xerox Metacode printers, you must have a GVG card. To print charts on IBM AFP printers, you must have a GOCA card.

The amount of hard disk space you will need depends mainly on the volume of data you must process. Keep in mind too, that the C/C++ compiler is only required if you plan to write your own custom rules and recompile the source modules provided in the Software Developer's Kit (SDK).

On UNIX Systems

For all UNIX systems, you can use any printer that supports IBM AFP, PCL, PostScript level 2, or Xerox Metacode. For HP printers, you need at least 6mb of memory, more if you are printing complicated graphics. The amount of hard disk space you need depends on the volume of data you process. Keep in mind too, that a compiler is only required if you plan to recompile the system, such as if you customize the source code or use a runtime library other than the one shown for your operating system.

NOTE For any UNIX installation, first make sure you have the uudecode, uncompress, and awk utilities installed.

On AIX systems

	Requirements
Operating system	AIX version 5.2 or higher
Model*	pSeries - Power RISC
Compiler	(Only necessary if adding custom code to the system) IBM Visual Age C/C+ version 6 IBM C/C++ Enterprise Edition for AIX v7
Runtime library	C Set ++ Runtime for AIX 5.0 or higher

	Tested on
Model*	pSeries p650
CPU	6 x 1.45GHz Power4+ processors
Memory*	12GB
Hard disk**	Two 36.4GB 10, 000RPM Ultra3 SCSI drives

*Additional memory and a faster CPU is not required, but will improve performance.

** Additional space required for your customized forms

On Linux systems

	Requirements
Operating system	GNU/Linux distributions
Model	Intel/AMD based systems
Compiler	(Only necessary if adding custom code to the system) GNU C/C++ compiler, gcc-3.2.3-58 or higher v3.2.3, gcc-c++-3.2.3-58 or higher v3.2.3
Runtime library	libgcc-3.2.3-58 or higher v3.2.3, libstdc++-3.2.3-58 or higher v3.2.3, compat-libstdc++-7.3-2.96.128 or higher v7.3, compat-glibc-7.x-2.2.4.32.6 or higher v7.x, glibc-2.3.2-95-30 or higher v2.3.2

	Tested on
Model*	Dell PWS450
Operating system	RedHat Enterprise Linux v3.1
CPU	2 x 2.40GHz Xeon processors
Memory*	3GB
Hard disk**	70GB SCSI drive

*Additional memory and a faster CPU is not required, but will improve performance.

** Additional space required for your customized forms

On Solaris systems

	Requirements
Operating system	Sun Solaris 9/SunOS 5.9 (SPARC based) or higher
Model*	UltraSPARC based
Compiler	(Only necessary if adding custom code to the system) Sun ONE Studio 8
Runtime library	Core Solaris 9

	Tested on
Model*	Sun Fire v240 Server
Operating system	Solaris v9 SPARC
CPU	2 x 1.28GHz UltraSPARC IIIi Cu
Memory*	2GB
Hard disk**	Four 36GB SCSI drives

*Additional memory and a faster CPU is not required, but will improve performance.

** Additional space required for your customized forms

On the PC Platform

The amount of hard disk space you will need depends mainly on the volume of data you must process. Keep in mind too, that a compiler is only required if you plan to recompile the system, such as if you customize the source code.

	Requirements
CPU	A 300 MHz or higher processor clock speed recommended (single or dual processor system); Intel Pentium/Celeron family, or AMD K6/Athlon/Duron family, or compatible processor recommended
Memory*	128mb RAM
Hard disk	1.5 gigabytes
Removable media	CD-ROM or DVD drive
Other components	Keyboard and mouse or compatible pointing device
Monitor	Color SVGA, 17-inch screen monitor
Printer	Any printer which supports PCL level 5 (HP IV or greater), PostScript level 2, IBM AFP, or Xerox Metacode
Printer memory**	8mb for HP printers
Compiler	Microsoft Visual Studio .NET 2003 for Windows XP Microsoft Visual Studio .NET 2003 for Windows 2000

* Additional memory, while not required, will improve system performance.

** Additional memory may be required if printing complicated graphics.

DOCUPRESENTMENT REQUIREMENTS

Your computer must have certain software and hardware components to run Docupresentment. Depending on your software license and operating environment, these requirements vary.

Docupresentment Workstation

For a Docupresentment workstation, you must have a personal computer equipped with the following:

- Microsoft Internet Explorer version 6.0 or later for Windows 2000 or Windows XP or higher
- Adobe • Acrobat Reader version 7.0 or higher

Docupresentment Server

You can run a Docupresentment on the following operating systems:

- Windows
- AIX
- Linux
- Solaris

NOTE For both Docupresentment Workstation and Server, you must have Java 1.5 or higher.

The following tables provide more detailed information on Skywire Software's minimum platform requirements for testing systems.

On Windows systems

	Tested on
Operating system	Windows 2000
CPU*	1xPentium II - 400 mHz
Memory*	256 MB
Hard disk (RTE)	25 MB
Hard disk (MRL)**	75 MB

*Additional memory and a faster CPU is not required, but will improve performance.

** Additional space required for your customized forms

On AIX systems

Requirements	
Operating system	AIX version 5.2 or higher
Model*	pSeries - Power RISC
Compiler	(Only necessary if adding custom code to the system) IBM Visual Age C/C+ version 6
Runtime library	C Set ++ Runtime for AIX 5.0 or higher
Tested on	
Model*	pSeries p650
CPU	6 x 1.45GHz Power4+ processors
Memory*	12GB
Hard disk**	Two 36.4GB 10, 000RPM Ultra3 SCSI drives
*Additional memory and a faster CPU is not required, but will improve performance.	
** Additional space required for your customized forms	

On Linux systems

Requirements	
Operating system	GNU/Linux distributions
Model	Intel/AMD based systems
Compiler	(Only necessary if adding custom code to the system) GNU C/C++ compiler, gcc-3.2.3-49 or higher, gcc-c++-3.2.3-49 or higher
Runtime library	glibc-2.3.2-95.30 or higher, compat-glibc-7.x-2.2.4.32.6, libstdc++-3.2.3-49 or higher, compat-libstdc++-7.3-2.96.128
Tested on	
Model*	Dell PWS450
Operating system	RedHat Enterprise Linux v3.1
CPU	2 x 2.40GHz Xeon processors
Memory*	3GB
Hard disk**	70GB SCSI drive
*Additional memory and a faster CPU is not required, but will improve performance.	
** Additional space required for your customized forms	

On Solaris systems

	Requirements
Operating system	Sun Solaris 9/SunOS 5.9 (SPARC based) or higher
Model*	UltraSPARC based
Compiler	(Only necessary if adding custom code to the system) Sun Workshop C/C++ v5.0
Runtime library	Core Solaris 9

	Tested on
Model*	Sun Fire v240 Server
Operating system	Solaris 9/SunOS 5.9 (SPARC based)
CPU	2 x 1.28GHz UltraSPARC IIIi Cu
Memory*	2GB
Hard disk**	Four 36GB SCSI drives

*Additional memory and a faster CPU is not required, but will improve performance.

** Additional space required for your customized forms

Web Server

This table outlines the web server requirements for each operating system:

Operating system Web server

Windows	2000 Server or 2003 Server (or higher), such as Microsoft Internet Information Server 4.0 (or higher).
AIX	Web server for AIX 5.2, such as IBM's HTTP Server for AIX version 1.3.3.1 or higher with the Java Runtime Environment and/or JDK for AIX, version 1.4.0 or higher.
Linux	Web server for Linux, such as Apache 1.3.12 or higher or IBM HTTP Server 1.3.9 or higher.
Solaris	Web server for Sun Solaris 7 or higher on SPARC, such as Java Web Server 2.0 or Apache 1.3.9 with the Java Runtime Environment and/or JDK for Solaris, version JRE 1.4.0 or higher. IBM HTTP Server 1.3.9 or higher can also be used.

NOTE IDS 2.0 requires Java 1.4 or higher. IDS 2.1 requires Java 1.5 or higher.

Skywire Software has tested Docupresentment and iDocumaker Workstation with the following web servers: Tomcat 4.0, Tomcat 5.0, WebSphere Application Server 4.0, and WebSphere Application Server 5.0

iPPS/iDOCUMAKER

Your computer must have certain software and hardware components to run iPPS or iDocumaker. This table outlines those requirements:

Basic requirements

	Requirements
CPU	Pentium III or greater
Operating systems	Windows 2000, Windows XP, or Windows 2003 Server
Memory*	256MB RAM
Hard disk**	400 MB free
Other components	Keyboard and mouse or compatible pointing device
Monitor	Color SVGA monitor

* Additional memory, while not required, will improve system performance.

** The amount of hard disk space you will need depends mainly on the volume of data you must process.

NOTE For iDocumaker Workstation, you must have Java 1.4 or higher.

Client requirements

In addition to the basic requirements, each client should have the following:

- Windows 2000 Professional or XP Professional or later
- Adobe Acrobat 7.0 or higher
- Microsoft Internet Explorer 6.0 or higher with these Internet security options enabled:
 - Run ActiveX controls and plug-ins
 - Script ActiveX controls safe for scripting
 - Allow cookies that are stored on your computer
 - Allow per-session cookies (not stored)
 - Java Permissions (set to Low Safety)
 - Access data sources across domains
 - Active Scripting
 - Scripting of Java Applets
 - Java console (requires restart)
 - Java logging
 - JIT compiler for virtual machine (requires a restart)
- Skywire Software's WIP Edit, version 11, patch 10 or higher
 - Java VM – Java console enabled (requires a restart)
 - Java VM – Java logging enabled
 - Java VM – JIT compiler for virtual machine enabled (requires a restart)

Server requirements In addition to the basic and client requirements, the computer you will use as a server should be configured with the following:

For COM+ implementations	For J2EE implementations
Internet Information Services (IIS) with World Wide Web Server and File Transfer Protocol (FTP) Server	Apache Tomcat or IBM WebSphere Note: Apache Tomcat has been tested but is not recommended for production use.
A database such as Microsoft Access database (Access 97 or higher), xBase database, or SQL database	A database such as Microsoft Access database (Access 97 or higher), xBase database, or SQL database
Visual Basic runtimes	not applicable
Microsoft XML Core Services 4.0 SP2 (msxml4)	not applicable
ADO 2.6 or later	not applicable
A static IP address	A static IP address

Web server For the web server, you should have:

- Minimum Pentium III with 512MB of RAM.
- Windows 2003 Server
- Component Services or Microsoft Transaction Server
- Microsoft Visual Basic 6 Runtimes (included/installed with iPPS 3.1)
- Microsoft Active Data Objects 2.6 or greater (included/installed with iPPS 3.1)
- IBM WebSphere MQ (formerly MQSeries) or Microsoft Message Queue client
- ODBC-compliant database (SQL Server/Oracle/DB2 recommended for production)
- Microsoft's XML parser MSXML 4.0 sp2
- Skywire Software's Docupresentment version 10.2 (IDS version 1.8) or greater

Recommendations **NOTE** Skywire Software has tested an iDocumaker Workstation J2EE version 3.1, patch 25 implementation on WebSphere Application Server (WAS) version 6.02 and WebSphere MQ Series version 6.0. We recommend you use WAS version 6.02 or greater.

We also recommend these additional products for your iPPS implementation:

Product	Description
FAP2HTML utility	Use to convert FAP files into HTML for iPPS. This utility is not needed with iPPS/iDocumaker com+ v3.11 or J2ee v3.1 because these products can dynamically produce HTML from a master resource library.
DHTML Text Editor	Use to alter forms.
Documaker Server	Use for rules publishing.

Using the right Java environment

This table shows various web servers Skywire Software has tested with and the Java version you should use with those web servers and with iPPS.

For	Use
Tomcat 4.0	Tomcat using Java 1.3 (iPPS using the J2EE standard 1.2)
Tomcat 5.0	Tomcat using Java 1.4 (iPPS using the J2EE standard 1.3)
WebSphere Application Server 4.0	WebSphere Application Server using Java 1.3 (iPPS using the J2EE standard 1.2)
WebSphere Application Server 5.0	WebSphere Application Server using Java 1.4 (iPPS using the J2EE standard 1.3)

Chapter 2

Installing Documaker Server on OS/390

This chapter provides information on how to install and configure Documaker on an OS/390 (MVS) system.

NOTE Following MVS version 5.2.2, new versions of this operating system were named OS/390. Following OS/390 version 2.10, new versions were named z/OS. Documaker Server runs on MVS, OS/390, and z/OS. In this manual, MVS, OS/390, and z/OS are referred to as OS/390 unless otherwise noted.

You will find information on these topics:

- [Overview on page 2](#)
- [Installing the System on page 4](#)
- [Upgrading Your System on page 6](#)
- [Customizing Your System on page 7](#)
- [Configuring Your Printer Type on page 11](#)
- [Configuring Host Archive Using DB2 on page 26](#)
- [Checking Your Installation on page 33](#)
- [Optimizing Performance on page 45](#)

In addition, you'll also find information about...

- [The LOADDAP Job on page 63](#)
- [OS/390 Libraries on page 69](#)
- [Uploading and Downloading Resources on page 72](#)
- [OS/390 Programs and Utilities on page 76](#)

OVERVIEW

This chapter provides you with steps to install and configure Documaker Server on your OS/390 system. This overview discusses the contents of this chapter.

NOTE See [System Requirements on page 2](#) for information about system requirements.

Installing the System

Documaker Server is distributed on 3480 cartridge tape for MVS systems. For detailed instructions, see [Installing the System on page 4](#). The process includes these steps.

- 1 Copy the LOADDAP job.
- 2 Modify and submit the LOADDAP job.

Configuring Your Printer Types

The system produces output for Xerox Metacode, IBM AFP, PostScript, PDF, or compatible printers. The steps differ, depending on the printer you will use.

Configuring your system for Xerox Metacode

Here is an overview of the steps required to configure the system for Xerox Metacode or compatible printers. For detailed instructions, see [Creating Xerox Metacode Printstreams on page 11](#).

- 1 Change the Xerox JSL.
- 2 Change the FSISYS file.
- 3 Pre-compile your FAP files.
- 4 Make your printer resources available. This can include uploading Xerox fonts, loading the fonts onto your printer, and converting logos.
- 5 Change the JCL for the DAP procedure (or the GenPrTX job) and the GENERXER job.

Configuring your system for AFP printers

Here is an overview of the steps required to configure the system for AFP or compatible printers. For detailed instructions, see [Creating AFP Printstreams on page 18](#).

- 1 Allocate AFP-related datasets.
- 2 Make your printer resources available. This can include uploading AFP fonts, loading page segments, setting up FormDefs, and creating overlays.
- 3 Pre-compile your FAP files into AFP overlays.
- 4 Modify the FSISYS file.
- 5 Modify the JCL for the DAP procedure (or the GenPrTX job) and the GENERAFP job.

Configuring your system for PostScript printers

Here is an overview of the steps required to configure the system for PostScript printers. For detailed instructions, see [Creating PostScript Printstreams on page 21](#).

- 1 Change the FSISYS INI file.
- 2 Modify the JCL.

Configuring your system to produce PDF files

Here is an overview of the steps required to configure the system to produce PDF files. For detailed instructions, see [Creating PDF Files on page 23](#).

- 1 Change the FSISYS INI file.
- 2 Modify the JCL.

Configuring Host Archive

Here is an overview of the steps required to configure host archive. This is an optional step. For detailed instructions, see [Configuring Host Archive on page 25](#).

Checking Your Installation

Here is an overview of the steps required to configure host archive. This is an optional step. For detailed instructions, see [Checking Your Installation on page 33](#).

- 1 Check your FSIUSER and FSISYS settings.
- 2 Check your JCL.
- 3 Run the DAPRUN procedure (or GenTrnX, GenDataX, and GenPrtX).
- 4 Run GenArcDX (or GenArcX).
- 5 Print the output.

Optimizing Performance

Once you have installed, configured, and tested your system, you can review this topic to find ways to improve system performance. For more information, see [Optimizing Performance on page 45](#).

Additional Information

At the end of this guide you will also find information about the LOADDAP job, MVS libraries, uploading and downloading resources, and MVS programs and utilities.

INSTALLING THE SYSTEM

For OS/390 systems, Documaker is distributed on a 3480 cartridge tape. The tape contains several files including JCL files, object modules, executable modules, and sample resources you can use to test your installation.

NOTE The object and executable modules were built using IBM's C/C++ for OS/390 version 2.10 compiler and corresponding Language Environment (LE) product.

COPYING THE LOADDAP Job

The LOADDAP job is included on the installation tape in the JCLLIB dataset. You can use the JCL listed below to unload the LOADDAP job. Refer to any documentation that came with the tape to verify the exact version number of the files and the volume serial number of the tape.

Make any necessary modifications to this JCL (jobcard, dataset names, and so on) and submit the job. This job should receive a return code **00**.

```
//JWCI      JOB (33005),'LOAD LOADJCL',CLASS=T,MSGCLASS=X,
//          NOTIFY=JWC
// * * * * *
// *          LOADLOAD - COPY THE LOADDAP JOB FROM TAPE TO MVS.
// *
// *
// * * * * *
//S1        EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//INDD      DD DSN=FSI.V110.JCLLIB,DISP=SHR,UNIT=CART,
//          VOL=SER=D11001
//OUTDD     DD DSN=FSI.V110.TEMP.JCLLIB,DISP=(,CATLG),UNIT=SYSDA,
//          SPACE=(CYL,(1,1,1))
//SYSIN     DD *
           C I=INDD,O=OUTDD
           SELECT MEMBER=LOADDAP
// *
```

MODIFYING THE LOADDAP JOB

Before submitting the LOADDAP Job, you should modify it to meet your system requirements.

Change any jobcard, dataset names, unit names, and so on so it will run on your system. The job uses an instream proc called LOAD. At the bottom of the job, you can specify the parameters that will be passed into the LOAD proc. This portion of the job is listed below along with a description of the parameters.

This job should receive a return code 00 or less in each step.

```
...  
  
//STEP1 EXEC LOAD,TUNIT=CART,  
//      TVOLSER=D11001,           <= CHANGE  
//      HLQ='FSI.V110',           <= CHANGE (seldom needed)  
//      CUSTHLQ='CUSTOMER.V110.INST', <= CHANGE  
//      DUNIT=SYSDA,  
//      PDSTYPE=LIBRARY
```

Parameter	Description
TUNIT	The unit name that should be used for this tape device.
TVOLSER	The tape volume serial number. The volume serial number is printed on the installation tape.
HLQ	The High Level Qualifier of the installation datasets. Normally, the value already specified for this parameter is correct.
CUSTHLQ	The High Level Qualifier that should be used when creating the installation datasets on your OS/390 system.
DUNIT	A generic or esoteric name for your OS/390 system's DASD units.
PDSTYPE	The type of PDS to create. By default, the LOADDAP job creates the installation partitioned datasets as PDS/Es, not PDSS. If your installation does not support PDS/Es or if you choose not to use PDS/Es, change the PDSTYPE parameter in the LOADDAP job from LIBRARY to PDS.

SUBMITTING THE LOADDAP JOB

The LOADDAP job reads the installation tape, creates the system datasets, sample resource datasets, font datasets, and so on and populates these datasets from the tape. LOADDAP should receive a return code 00 in every step.

The LOADDAP job is listed on [page 63](#)

NOTE Once the LOADDAP job is finished, the JCLLIB will contain a sample JOBCARD member which can be copied into and used as a jobcard in subsequent jobs.

UPGRADING YOUR SYSTEM

Upgrading to a new version of Documaker or applying corrective service (patches) to your Documaker system requires that you perform a standard Documaker installation from tape.

DETERMINING YOUR SYSTEM'S PATCH LEVEL

You can determine the patch level of your Documaker system by running a Job provided in the FSIVERX member of JCLLIB. This Job (see below) runs the FSIVER utility which examines the contents of ~~LINKLIB~~ dataset and looks for patch identifiers contained in the Documaker program modules in the LINKLIB dataset.

The FSIVER utility then produces a report listing the patches it found. Unless instructed otherwise, FSIVER produces both a detailed patch report and a summary patch report. See the [Docutoolbox Reference](#) for more information on the FSIVER utility.

```
//ZDA      JOB (33005),'FSIVER - 110 ',CLASS=T,MSGCLASS=X,
//          NOTIFY=&SYSUID
//*
//          SET HLQ='FSI.V110' <== SET HIGH LEVEL QUALIFIER
//          SET RES='RPEX1'      <== SET RESOURCE (E.G. RPEX1, UTEX1)
//*
//          JCLLIB ORDER=&HLQ..PROCLIB
//
//
*****
/* PROGRAM : FSIVER
/* PURPOSE : CREATES A REPORT THAT LISTS WHICH PATCHES HAVE BEEN
/*           APPLIED TO THE PROGRAMS IN THE LINKLIB REFERENCED BY
/*           THE LINKLIB DD STATEMENT.
/*
/* PARS      : /I=PROGRAM (NAME OF MEMBER IN DD:LINKLIB)
/*           OR '**' TO LIST PATCH LEVEL OF ALL PROGRAMS IN
/*           DD:LINKLIB.
/*
//
*****
//FSIVER EXEC PGM=FSIVER,PARM='/ /I=*'
//
//STEPLIB DD DSN=&HLQ..LINKLIB,DISP=SHR
//          DD DSN=SYS1.SCEERUN,DISP=SHR
//LINKLIB DD DSN=&HLQ..LINKLIB,DISP=SHR
//SYSPRINT DD SYSOUT=*
```

CUSTOMIZING YOUR SYSTEM

This step is optional.

Documaker is designed as a rules-based engine. Many rules are provided in the base product. These rules are located in the various source modules that make up a segment of the product called RULLIB. The Documaker architecture also lets you add your own rules, called *customrules*. Some custom rules are provided, as examples, in a segment of the product called CUSLIB. If you want to modify these custom rules or add custom rules of your own, you should make your changes or additions to the CUSxxxxx modules that are provided in the Documaker Software Developer's Kit (SDK).

After making changes to any source modules in the SDK, you will need to compile these source modules into object modules and then link a new GenData program. The GenData program is also known as the Rules Processor.

IF YOU DO NOT HAVE A SOURCE CODE LICENSE

The standard Documaker license includes a subset of the Documaker source and header modules referred to as the Software Developer's Kit (SDK). On OS/390, this SDK comprises the SOURCE and INCLIB datasets. Follow the steps outlined in the following topics.

Running the COMPSDK Job

To compile the modules provided in the Documaker SDK, modify and submit the job in COMPSDK. The COMPSDK job invokes the CBC compile proc for approximately 56 source modules. Each source module is compiled into an object module, and the object modules are written to the OBJLIB. The COMPSDK job should receive a return code of 04 or less in each step. A portion of the COMPSDK job is shown below.

```

/**** COPY JOBCARD HERE...
/*
/* COMPSDK - COMPILE JCL FOR THE DOCUMAKER SOFTWARE DEVELOPER KIT
/*          (SDK), WHICH IS A SUBSET OF THE COMPLETE DOCUMAKER
/*          SOURCE.
/*
/*CUSARC EXEC CBC, MEM=CUSARC
/*CUSBANNR EXEC CBC, MEM=CUSBANNR
/*CUSBAT EXEC CBC, MEM=CUSBAT
/*CUSBITMP EXEC CBC, MEM=CUSBITMP
/*CUSCALLB EXEC CBC, MEM=CUSCALLB
...

```

Running the GENDATAL Job

You can link-edit a new GenData program to incorporate any changes or additions you have made to the CUSxxxxx source modules. Run the job in JCLLIB member GENDATAL to link-edit GenData. The GENDATAL job first invokes the OBJUTIL proc to rebuild the object library directory, then it invokes the LINK proc to link-edit GenData. GENDATAL should receive a return code of 04 or less in each step. The GENDATAL job is shown below.

```
/* COPY JOBCARD HERE
/* * * * * *
/*
/*          LINK-EDIT GENDATA
/*
/* * * * * *
/*
//ALIAS      EXEC OBJUTIL
//
//GENDATA EXEC LINK, MEM='GENDATA'
```

Running the GENPRTL Job

You can link-edit a new GenPrint program to incorporate any changes or additions you have made to the CUSxxxxx source modules. Run the job in JCLLIB member GENPRTL to link-edit GenPrint. The GENPRTL job first invokes the OBJUTIL proc to rebuild the object library directory, then it invokes the LINK proc to link-edit GenPrint. GENPRTL should receive a return code of 04 or less in each step. The GENPRTL job is shown below.

```
/* COPY JOBCARD HERE
//
*****
/*
/*          LINK-EDIT GENPRINT
/*
//
*****
/*
//ALIAS      EXEC OBJUTIL
//
//GENPRINT EXEC LINK, MEM=GENPRINT
```

NOTE If you purchased a license for PDF or PDF417 or both, the object modules you received include these features and will enable the resulting GenData and GenPrint to have the capabilities you licensed.

IF YOU HAVE A SOURCE CODE LICENSE

Follow the steps outlined in the following topics.

Running the COMPALL Job

This task is optional. If you do have a source code license, you can compile the full source using the COMPALL member in JCLLIB. The COMPALL member invokes one of the compile procs, which can be found in PROCLIB, for each of the 950 or so source members. You will usually need to separate the COMPALL member into several jobs. COMPALL compiles each source module into an object module. The object modules are written to the OBJLIB. The COMPALL job should receive a return code of 4 or less in each step.

A portion of the COMPALL Job is shown below.

```
//*** COPY JOBCARD HERE...  
//ADLER32A EXEC CBC, MEM=ADLER32A  
//AFPAEG EXEC CBC, MEM=AFPAEG  
//AFPA2E EXEC CBC, MEM=AFPA2E  
//AFPBAG EXEC CBC, MEM=AFPBAG  
//AFPBARCD EXEC CBC, MEM=AFPBARCD  
...
```

Running the LINKALL Job

This task is optional. Once the source modules are compiled into object modules, you need to link-edit the Documaker programs and utilities by running the LINKALL job. The LINKALL job first invokes the OBJUTIL proc, which rebuilds the object library directory, then it invokes the LINK proc for each program.

The LINKALL job should receive a return code of 0 or less in each step. A portion of the LINKALL job is shown below.

```
/* COPY JOBCARD HERE ...
//
*****
/*      LINKALL - LINK ALL EXECUTABLES.
/*
/*      1. RUNS THE OBJECT UTILITY TO REBUILD THE @@DC370$ MEMBER
/*      OF THE OBJECT LIBRARY.
/*
/*      2. LINK-EDIT ALL OF THE EXECUTABLES FOR DAP.
/*
//
*****
/*
//ALIAS1 EXEC OBJUTIL
/*
//AFP2MVS EXEC LINK, MEM=AFP2MVS      <== USED FOR AFP PRINT
//AFPCOPY EXEC LINK, MEM=AFPCOPY
//AFPMDDEF EXEC LINK, MEM=AFPMDDEF    <== USED FOR AFP PRINT
//BARR2MVS EXEC LINK, MEM=BARR2MVS    <== USED FOR XEROX / BARR
//BARRWRAP EXEC LINK, MEM=BARRWRAP    <== USED FOR XEROX / BARR
...

```

CONFIGURING YOUR PRINTER TYPE

On MVS, you can use the system to create print streams for these printing environments:

- Xerox Metacode (see [Creating Xerox Metacode Printstreams below](#))
- AFP (see [Creating AFP Printstreams on page 18](#))
- PostScript (see [Creating PostScript Printstreams on page 21](#))
- PDF (see [Creating PDF Files on page 23](#))

The following topics tell you how to configure your system for these environments.

CREATING XEROX METACODE PRINTSTREAMS

Configuring your system to produce Xerox Metacode print streams involves these tasks:

- Making changes to the Xerox JSL
- Making changes to the FSISYS INI file
- Pre-compiling your FAP files into MET files (this task is optional)
- Making printer resources available
- Making modifications to the JCL

Changing the Xerox JSL

Configuring the system to print to a Metacode printer involves setting various initialization file settings, specifically those in the PrtType:XER control group in the FSISYS initialization file, which is usually a member of the DEFLIB PDS.

To know what values need to be used for each setting, look at the Xerox printer's configuration file. Xerox configuration settings are kept on the printer in a JSL file. The JSL file contains the initial settings under which the printer operates.

You can have multiple JSL files resident on a single printer, with the desired JSL file specified when you start the printer. It is not necessary to start the printer with a specific JDE,JSL combination to print Metacode streams. You must, however, have a JDE,JSL combination which can be dynamically switched to for the duration of that print stream that satisfies the requirements explained below.

Specific values from the JDE,JSL combination must be available to the GenPrint program. You make these values available by placing them in the FSISYS file.

Viewing the JSL

Before you look at any JSL files, first determine which JSL file will be active when print files are sent to the printer. The JSL file name can contain up to six characters and always has a JSL extension. Most Xerox Metacode printers (except the 4235) contain a built-in console editor for creating, viewing, modifying, and printing JSL files.

NOTE You may want to print your Xerox printer's current JSL file before you continue.

JSL requirements To print a Metacode stream on your Xerox printer, you must have a Job Description Entry (JDE) with the parameter `VOLUME CODE=NONE` in your JSL's default JDE entry or any other JDE entry in the JSL contains this parameter, you do not need to add another JDE entry. If the parameter is missing, you must add it. Follow your accepted procedures for making such JSL additions and modifications.

Once there is a JDE entry (or the default entry) that satisfies the `VOLUME CODE` requirement, enter values in the table below that correspond either to this JDE entry specifically or to the JSL file in general.

Use the table below to note the values in the Xerox JSL you will use. You will enter these values when you modify the FSISYS file.

JSL entry	Description	Values
DJDEI	Corresponds to the IDEN statement's PREFIX= keyword (or PRE=). Indicates E for EBCDIC, A for ASCII, X for hexadecimal and a string value. If there is no E,A or X before the string value, an E is assumed by default.	
DJDEO	Corresponds to the IDEN statement's OFF= keyword.	
DJDES	Corresponds to the IDEN statement's SKIP= keyword.	
ImageOpt	If the JSL contains a "GRAPHICS = YES" statement, set ImageOpt to Yes. Otherwise, set it to No. You must set this option to Yes if you need the system to dynamically draw charts.	
JDEName	If you had to create a new JDE entry to specify <code>VOLUME CODE = NONE</code> , use the name of the new JDE entry. If a JDE entry already existed that contained <code>VOLUME CODE = NONE</code> , specify the name of that JDE (if it's the default, it's probably called <i>DFLT</i>).	
JDLName	The name of the JSL file.	
JDLHost	Corresponds to the VOLUME parameter's HOST= keyword. This value will be either IBMONL or IBMOS. If not specified, the default is <i>IBMOS</i> .	
JDLCode	Corresponds to the VOLUME statement's CODE= keyword for the initial JSL setting (at the top of the JSL file). This value will be either EBCDIC, ASCII, or NONE.	

JSL entry	Description	Values
JDLData	Corresponds to the LINE statement's DATA= keyword. Should be two numbers separated by a comma (omit the parentheses).	
PrinterInk	If the printer supports a highlight color, specify that color here. Otherwise, leave blank.	
JDLRStack	To find this value, find the RSTACK command (if there is one) and work backwards. The RSTACK command refers to a TEST label, which sets up a string test and refers to a CONSTANT, string, or value. For example... RSTACK: TEST=RST,DELIMITER=YES; RST: CRITERIA CONSTANT =(0,10,EQ,RSTK); RSTK: TABLE CONSTANT =X'13131313131313131313'; ...the value you would use would be: JDLRStack=0,10,EQ,X'1313131313131313131313131313'	
JDLROffset	Locate the ROFFSET command (if there is one) and work backwards. See JDLRStack above.	
JDLRPage	Locate the RPAGE command (if there is one) and work backwards. See JDLRStack above.	

Changing the FSISYS INI File

NOTE Two sets of resources are included on the installation tape. The RPEX1 set of resources is designed for the insurance market. The UTEX1 resources are designed for the utility market. You can verify your installation by running the system using one or both of these sets of resources.

The following instructions may ask you to modify the FSISYS initialization member of DEFLIB, so the DEFLIB you must modify is the one for the set of resources you plan to use, such as *hlq.v103.RPEX1.DEFLIB* or *hlq.v103.UTEX1.DEFLIB*, both if you plan to use both.

Metacode data sets on MVS should be created with a Variable Blocked Machine character (VBM) Record Format (RECFM) and should have a Logical Record Length (LRECL) that is four bytes longer than the length of the longest Metacode record you plan to write.

MVS uses four bytes of each record in a Variable Blocked data set to store data about that record. For example, in your INI file you may have the JDLData option set to

0,250

which indicates the longest Metacode record will be 250 bytes long. If this is the case, your Metacode data set should be allocated with an LRECL of 254.

PrtType:XER control group

You must place the values you gathered from or added to the Xerox printer's JSL file into the PrtType:XER control group in the FSISYS INI file. This file is generally stored as a member called **FS/SYS** within the sample DEFLIB PDS.

Below are the options in the PrtType:XER control group relevant to MVS environments. Using the table you completed earlier, make changes as necessary to the following values. The example below shows the default FSISYS values.

NOTE You can comment out an INI control group or option by placing a semi-colon (;) before the control group or option.

```
< PrtType:XER >
  ColorCharts           = Yes
  CompileInstream       = No
  DownloadFonts         = No, Disabled
  InitFunc              = XERInit
  Module                = XERW32
  OutputFunc            = XEROutput
  OutMetFunc            = XEROutMet
  PageNumbers           = Yes
  PrintFunc             = XERPrint
  PrinterInk            = Blue
  TermFunc              = XERTerm
  SendColor             = Yes
  DJDEIden              = A'@@@DJDE'
  DJDEOffset            = 0
  DJDESkip              = 8
  Environment           = MVS
  OutMode               = JES2
  ImageOpt              = Yes
  JDEName               = DFLT
  JDLData               = 0, 255
  JDLHost               = IBMONL
  JDLName               = DFAULT
  JDLCode               = NONE
  JDLRStack             = 0, 10, EQ, X'13131313131313131313'
  JDLRPage              = 0, 5, EQ, X'FFFF26FFFF'Printer control group
```

The Printer control group tells the GenPrint program what type of output to produce. For the RPEX1 sample resources, the PrtType option is ~~XER~~ **AFP**. For the UTEX1 sample resources, the PrtType option is set ~~to AFP~~. Make sure the option is set correctly for your environment. For example, this setting tells the GenPrint program to produce Xerox Metacode print streams:

```
< Printer >
  ;PrtType = AFP
  PrtType = XER
```

Printer1 control group The Printer1 control group contains PORT options which specify a DD name to which the system writes output for that recipient batch.

```
< Printer1 >  
PORT = DD:PRTBAT1
```

There may be other options present but they will be commented out with a semi-colon. Make sure the PORT option is set appropriately for your environment.

Pre-compiling FAP Files into Metacode Files

During the GenPrint process, FAP files (images) are compiled into Metacode files and arranged into the print stream. To speed execution, you can pre-compile—compile before running GenPrint—these FAP files.

NOTE The FISISYS file included with the sample resources assumes you will be using pre-compiled Metacode files, *oPMETs* since the use of PMETs increases performance. If, however, you *do not* want to run with PMETs, change the CompileInstream option in the PrtType:XER control group and DownloadFAP INI option in the RunMode control group to *Yes* and skip to the topic [Making Printer Resources Available on page .17](#)

You can use the FAP2MET utility to compile FAP files into Metacode files. To run the system using pre-compiled Metacode files you must run the FAP2MET utility for every FAP file in FORMLIB. Here's how:

- 1 Run the FAP2METS job in JCLLIB. This job deletes or re-allocates the PMETLIB PDS and runs the FAP2MET utility for every FAP file in FORMLIB or FAPLIB. This creates a corresponding *PMET* and stores it in the PMETLIB PDS. The FAP2MET job should receive a return code of 00 for each step.
- 2 Make sure the CompileInstream option in the PrtType:XER control group is set to *No* and that the DownLoadFAP option in the RunMode control group is set to *No*.
- 3 Modify the JCL for the DAP procedure or the GenPrtX job (please first read the topic [Changing the JCL on page 16](#)) so the PMETLIB DD statement is uncommented (it is included in the JCL but is commented out). This lets the GenPrint program open and read the PMETs you created.

Making Printer Resources Available

The resources your Xerox Metacode printer needs fall into two categories:

- Fonts
- Logos

The following topics tell you how to make these resources available to your printer.

Fonts By default, the verification test (see [Checking Your Installation on page 33](#)) uses the AGFA fonts included with the system. If you installed the system from tape, the fonts were loaded into datasets called &HLQ..AGFA.XEROX.FONTLIB and &HLQ..AGFA.XEROX.ROTATED.FONTLIB when you ran the LOADDAP job. If that's the case, go to the topic [Loading fonts onto your printer on page 17](#)

Loading fonts onto your printer

To load the AGFA fonts onto a Xerox printer, follow these steps:

NOTE The names of the Xerox fonts begin with the character **X**. You may want to list the existing fonts on the printer to see whether any of your existing fonts have the same names as the fonts you are about to send to the printer.

- 1 Run the XERDNLDS member in JCLLIB. This job uses the FISISYS file you customized earlier. It converts the fonts into a format your Xerox printer can store. XERDNLDS should complete all steps with a return code of 00.
- 2 Run the GENDNLDS member in JCLLIB. This job runs the IBM utility IEBGENER to copy each font to the Xerox printer. Modify GENDNLDS as necessary to fit your standards. Be sure to set the SYSOUT output class to the JES2 output class for the Xerox printer. GENDNLDS should complete all steps with a return code of 00.

Logos The verification test references FAP files which contain logos (bitmap-type graphics). You must place these logos on the Xerox printer either as FNT files or as IMG files.

If your Xerox printer has a graphics card, it can print IMG files and you should place the LOGOs on the printer as IMG files. If your printer does not have a graphics card, it cannot print IMG files—place the logos on the printer as FNT files.

When you installed the system, two PDSs were created called &HLQ..LOG2IMG and &HLQ..LOG2XFNT. These PDSs contain logos which have been converted into IMG and FNT files, respectively. You should download the contents of one of these PDSs to your Xerox printer.

For steps 1 and 2, choose either to load the IMG files (choose DD statement for LOG2IMG) or the FNT files (choose DD statement LOG2XFNT) to your Xerox printer:

- 1 Run the XERLOGOS job in JCLLIB. Edit the job to read from and write to the appropriate type of file (LOG2IMG or LOG2XFNT). This job should complete with return codes of 00 in all steps.
- 2 Run the GENLOGOS job in JCLLIB. Edit the job so it will read from the dataset you created in step 1 and so it will send the output to the correct JES sysout class. This makes sure the Xerox resources are sent to the Xerox printer. This job should complete with a completion code of 00 in all steps.

Changing the JCL

DAP procedure (or the GenPrtX job)

In the next topic you will run the system with the sample resources to make sure it has been installed and configured correctly. You must modify the jobs you will run to conform to your local JCL and environment standards. The last job, GenPrint, will reference the DD names you set up in the topic [Printer1 control group on page 16](#)

When you run the system with the sample resources, you can run each step as a separate job (GenTrnX, GenDataX, and GenPrintX) or in sequence together by running a job called DAPRUN. The DAPRUN job invokes the DAP procedure which executes GenTrn, GenData, and GenPrint in this order.

For the verification test, you should use the DAPRUN job and edit the DAP procedure in PROCLIB. If, however, you want to run each step separately, edit the GenPrtX member in JCLLIB.

GENERXER job

You use the GENERXER job in JCLLIB to send the print streams created by the GenPrint program to the JES spool to be printed by your Xerox printer. Modify this job as necessary to match the file names you have used.

CREATING AFP PRINTSTREAMS

Configuring your system to produce AFP print streams involves these tasks:

- Allocating AFP-related datasets
- Making printer resources available.
- Pre-compiling FAP files into overlays (this task is optional)
- Making changes to the FSISYS initialization file
- Making changes to the JCL

Allocating AFP-related Datasets

To allocate the datasets required to perform AFP printing, use the ALLOCAFP job in the JCLLIB member. This job allocates the Form Definition Library (FDEFLIB), the Page Segment Library (PSEGLIB), and the Overlay Library (OVERLIB). The job should complete with a return code of 00.

Making Printer Resources Available

The resources your IBM AFP printer needs fall into these categories:

- Fonts
- Page segments
- FormDefs
- Overlays

The following topics tell you how to make these resources available to your printer.

Fonts By default, the verification test (see [Checking Your Installation on page 33](#)) uses the Agfa fonts included with the system. If you will print using IBM's Advanced Function Presentation (AFP), provided by the IBM Print Services Facility (PSF), you need to use these Agfa fonts.

The installation tape includes the Agfa 240-dpi fonts. If you want the full compliment of Agfa fonts, which includes 300-dpi fonts, contact Support Services and ask for the Agfa Font CD.

Page segments Logo Manager generates bitmap-type graphics in ~~logo~~ files which, on the PC, have an extension of *.LOG. The verification test uses images that contain these logos. For these images to print correctly, you must convert these logos into AFP Page Segments (PSEGS). You can run the LOG2PSEG utility on either the workstation or the MVS system to perform this task.

To convert logos for the verification test, run the LOG2PSGS job in JCLLIB. This job converts several logos from the sample FORMLIB or FAPLIB into page segments and places them into a PSEGLIB. The job should complete all steps with a return code of

FormDefs Next, submit the AFPFMDFX job in JCLLIB to compile ~~source~~ Form Definition File into an AFP-ready FormDef. This job should complete with a return code of

Overlays To speed performance in the GenPrint process, you can pre-compile FAP files (images) into AFP overlays before you run the GenPrint program. Follow the directions in the next topic, Pre-compiling FAP Files into AFP Overlays, to create the AFP overlays.

NOTE The FSISYS file included with the sample resources assumes you will be using AFP overlays, since the use of overlays increases performance. If, however, you do not want to run with overlays, change the SendOverlays option ~~No~~ and the DownLoadFAP option to ~~Yes~~ and skip to the topic [Changing the FSISYS INI File on page 19](#)

Pre-compiling FAP Files into AFP Overlays

The FAP2OVL utility pre-compiles FAP files into AFP overlays, which can improve performance. To pre-compile FAP files into AFP overlays, run the FAP2OVLS job in JCLLIB. This job converts the FAP files in the sample FAPLIB PDS into overlays and places them into the OVERLIB PDS.

Changing the FSISYS INI File

PrtType:AFP control group

Here are the options in the PrtType:AFP control group in the FSISYS file which apply to MVS environments. The FSISYS INI file should have been supplied with the value below:

```
< PrtType:AFP >
  SendOverlays=Yes
```

If you chose not to create AFP overlays (in the prior topic above), set the SendOverlays option to *No*. If you created overlays, make sure the option is set ~~Yes~~

Printer control group The Printer control group tells the GenPrint program what type of output to produce. For the RPEX1 sample resources, the PrtType option is set to ~~XER~~ **AFP**. For the UTEX1 sample resources, the PrtType option is set to ~~AFP~~ **XER**. Make sure the option is set correctly for your environment. For example, this setting tells the GenPrint program to produce AFP print streams:

```
< Printer >
  PrtType = AFP
;PrtType = XER
```

Printer1 control group The Printer1 control group contains PORT options which specify a DD name to which the system writes output for that recipient batch.

```
< Printer1 >
  PORT = DD:PRTBAT1
```

There may be other options present but they will be commented out with a semi-colon. Make sure the PORT option is set appropriately for your environment.

Changing the JCL

**Using the DAP procedure
(or the GenPrtX job)**

In the topic, [Checking Your Installation on page 33](#), you will run the system using the sample resources. This will help you make sure your system is installed and configured correctly. Before you run the test, you must modify the jobs you will use during the test to match your local JCL and environment standards. The last job, GenPrint, references the DD names you may have added or uncommented ([See Type:AFP control group on page 19](#)).

When you run the system with the sample resources, you can run each step as a separate job (GenTrnX, GenDataX, and GenPrintX) or in sequence together by running the DAPRUN job. The DAPRUN job invokes the DAP procedure which executes GenTrn, GenData, and GenPrint in this order.

For the verification test, we recommend you use the DAPRUN job. If you choose to use the DAPRUN job, edit the DAP procedure in PROCLIB. If, however, you want to run each step separately, edit the GenPrtX member in JCLLIB.

NOTE The DAP procedure included in PROCLIB (or the GenPrtX member of JCLLIB) for the UTEX1 resources is set up to produce AFP output file names such as AFPBAT1, AFPBAT2, and so on. You may want to inspect the procedure to make sure these names are correct, but you should not have to change the names.

GENERAFF job You will use the GENERAFP job in JCLLIB to send the print streams created by the GenPrint program to the JES spool to be printed by your AFP printer. Modify this job as necessary to match the file names you have used.

CREATING POSTSCRIPT PRINTSTREAMS

Configuring the system to produce PostScript printstreams involves these tasks:

- Making changes to the FSISYS INI file
- Making changes to the JCL
- Printing the PostScript printstream

Changing the FSISYS INI File

Here are the options in the PrtType:PST control group in the FSISYS INI file that are relevant to MVS environments:

```
< PrtType:PST >
  Module           = PSTW32
  PageNumbers      = Yes
  Printfunc        = PSTPrint
; PrinterModel     = (PPD file)
  SendOverlays     = Yes
  SendColor        = Yes
  DownloadFonts    = Yes
```

If you are using a PostScript Printer Description (PPD) file, upload your PPD file (as text) to DEFLIB, uncomment the PrinterModel option by removing the semicolon (;), and set it to your PPD file in DEFLIB. For example, if your PPD file in DEFLIB is HP8000, set your option as shown here:

```
PrinterModel      = HP8000
```

Setting the SendOverlays option to Yes tells the GenPrint program to use PostScript overlays to speed its process. To create PostScript overlays, first run the OVLCOMP utility on Windows to convert your FAP files into PostScript overlays (the OVLCOMP utility only runs on Windows). You then upload (as binary) the PostScript overlays to the dataset referred to by OVERLIB DD in your JCL.

The Printer control group tells the GenPrint program what type of output to produce. For the RPXE1 sample resources, the PrtType option is set to XER. For the UTEX1 sample resources, the PrtType option is set to AFP. Make sure the option is set correctly for your environment. For example, this setting tells the GenPrint program to produce PostScript print streams:

```
< Printer >
;PrtType = AFP
PrtType = PST
;PrtType = XER
;PrtType = PDF
;EnableTransBanner      = True
;EnableBatchBanner      = True
;BatchBannerBeginScript = PreBatch
;TransBannerBeginScript = PreTrans
;BatchBannerEndScript   = PstBatch
;TransBannerEndScript   = PstTrans
```

The Printer1 control group contains PORT options which specify a DD name to which the system writes output for that recipient batch.

```
< Printer1 >  
    PORT = DD:PRTBAT1
```

There may be other options present but they will be commented out with a semicolon. Make sure the PORT option is set appropriately for your environment.

Changing the JCL

In the next topic, Check Your Installation XX, you run the system with sample resources to make sure it has been installed and configured correctly. You must modify the jobs you will run to conform to your local JCL and environment standards.

The last job, GenPrint, references the DD names you set up in the Printer1 control group.

When you run the system with the sample resources, you can run each step as a separate job (GenTrnX, GenDataX, and GenPrtX) or in sequence together by running a job called DAPRUN. The DAPRUN job invokes the DAP procedure which executes GenTrn, GenData, and GenPrint in this order.

For the verification test, use the DAPRUN job and edit the DAP procedure in PROCLIB. If, however, you want to run each step separately, edit the GenPrtX member in JCLLIB.

Printing PostScript Printstreams

To print PostScript print streams to a PostScript-capable printer, download (as binary) the print streams created by the GenPrint program to your Windows environment, and do a binary copy (see below) of the downloaded file to your PostScript printer.

```
>copy /b your_file your_printer
```

CREATING PDF FILES

NOTE Creating PDF files from an OS/390 installation requires additional licensing. Contact your sales representative for additional information.

Configuring the system to produce PDF prints involves these tasks:

- Making changes to the FSISYS INI file
- Making changes to the JCL
- Creating the PDF file

Changing the FSISYS INI File

In the FSISYS INI file, below are the options in the PrtType:PDF control group relevant to MVS environments. The FSISYS INI file includes the values shown here:

```
< PrtType:PDF >
; Device           = E:\TEST.PDF
  BookMark         = Yes,Page
  DownloadFonts    = Yes
  LanguageLevel    = Level1
  Module           = PDFW32
  PageNumbers      = Yes
  PrintFunc        = PDFPrint
  SendOverlays     = No
  SendColor        = Yes
  Encrypt          = N
  SecurityGroup    = PDFEncryption
; PrintViewOnly    = No
; SplitText        = No
; SplitPercent     = 50
; Class           = PDF
; PrePrintedPaper= (not supported)
; PaperSize       = 0
  Compression      = 0
  FontCompression= 0
< PDFEncryption >
  KeyLength        = 128
  User             = M
  Owner            = Docucorp
  AllowModify      = FALSE
```

To produce secured PDF printstreams, set Encrypt option to Yes. Under the PDFEncryption control group, set the KeyLength, User, and Owner options as appropriate for your implementation.

The Printer control group tells the GenPrint program what type of output to produce. For the RPEX1 sample resources, the PrtType option is set to XER. For the UTEX1 sample resources, the PrtType option is set to AFP. Make sure the option is set correctly for your environment.

For example, this setting tells the GenPrint program to produce PDF printstreams:

```
< Printer >
;PrtType = AFP
;PrtType = PST
;PrtType = XER
PrtType = PDF
EnableTransBanner      = True
EnableBatchBanner      = True
BatchBannerBeginScript = PreBatch
TransBannerBeginScript = PreTrans
BatchBannerEndScript   = PstBatch
TransBannerEndScript   = PstTrans
```

The Printer1 control group contains PORT options which specify a DD name to which the system writes output for that recipient batch.

```
< Printer1 >
PORT = DD:PDFLIB(BAT10000)
```

There may be other options present. If so, preface them with a semicolon to comment them out. Make sure the PORT option is set appropriately for your environment.

In the RunMode control group, you will see these values:

```
< RunMode >
NAUnload      = Yes
DownloadFAP   = No
CompiledFAP   = Yes
ChkDstLenExceeded = No
LoadCordFAP   = No
; MergedDDTs  = No
```

Set the DownloadFAP option to Yes when producing PDF.

Changing the JCL

In the topic, [Checking Your Installation on page 39](#), you will run the system with sample resources to make sure it has been installed and configured correctly. You must modify the jobs you will run to conform to your local JCL and environment standards.

The last job, GenPrint, references the DD names you set up in the Printer1 control group.

When you run the system with the sample resources, you can run each step as a separate job (GenTrnX, GenDataX, and GenPrtX) or in sequence together by running a job called DAPRUN. The DAPRUN job invokes the DAP procedure which executes GenTrn, GenData, and GenPrint in this order.

For the verification test, use the DAPRUN job and edit the DAP procedure in PROCLIB. If, however, you want to run each step separately, edit the GenPrtX member in JCLLIB.

Creating the PDF Printstream

To print PDF printstreams to the printer, download (as binary) the print streams created by the GenPrint program to your Windows environment, and print to the printer by using your Adobe Reader to open the downloaded file and print.

CONFIGURING HOSTARCHIVE

The system lets you archive form sets that have been created by the GenData program. Archiving is performed by running the GenArc program and can be configured to archive the form sets in one of several ways:

- To the standard DAP archive. For more information, see [Configuring Host Archive Using DB2 on page 26](#)
- To Documange. For more information, see [Archiving to Documange on page 30](#)
- To Stacked DPA files. For more information, see [Archiving Using Stacked DPA on page 32](#)

CONFIGURING HOST ARCHIVE USING DB2

In addition to (or instead of) simply printing form sets, you can optionally archive your form sets and data using the GenArc program. The GenArc program on MVS systems supports archival to IBM's DB2 database management system.

Configuring a host archive system involves performing these tasks:

- Creating DB2 objects
- Running BINDDB2 or the ALLOCDB2 and COMPDB2 jobs
- Modifying the FSISYS file
- Making JCL changes

Creating DB2 Objects

The GenArc program, as configured for the sample library, uses these DB2 tables:

Table	Description
APPIDX	An application index table which contains a row for every archived form set.
ARCHIVE	An archive table which contains the actual NAFILE and POLFILE data for each archived form set.
CATALOG	A table which stores unique catalog IDs which the system uses as it builds the ARCKEY used in the APPIDX and ARCHIVE tables.
RESTART	A table in which GenArc restart information is stored.

You should create these tables using the ARCSQLR Job for the RPEX1 resources and the ARCSQLU Job for the UTEX1 resources. Modify the Jobs to suit your site's DB2 naming conventions and submit. The job should receive a return code of 00.

Running the BINDDB2 (or COMPDB2) Jobs

If you have a standard license, run the BINDDB2 job in JCLLIB as described below. If you have a source code license, skip to the following topic, [Running ALLOCDB2 and COMPDB2 on page 27](#)

The BINDDB2 Job performs DB2 binds for the DB2LIB and DB2LIB2 packages and the DAP110 plan. In the BINDDB2 job, change the DB2 subsystem, package names, and plan name to fit your standards. The BINDDB2 job should finish with return code 00 in all steps. Proceed to [Modifying the FSISYS INI File for Host Archive on page 27](#)

Running ALLOCDB2 and COMPDB2

The COMPDB2 job invokes the DB2 pre-compiler to pre-compile source files and place the resulting DBRMs into a PDS called DBRMLIB. If you installed Documaker from tape, this DBRMLIB dataset should exist, but if it is not available, you can allocate it by running the job ALLOCDB2. The ALLOCDB2 job should finish with a return code of 00.

The GenArc program uses programs called DB2LIB and DB2LIB2 to make all SQL calls. These SQL calls are prepared dynamically and are not static SQL calls.

To prepare the GenArc program for execution, run the job in member COMPDB2 of JCLLIB. The COMPDB2 member uses the DSNHC procedure in PROCLIB. Essentially, the COMPDB2 job will...

- DB2- pre-compile the DB2LIB and DB2LIB2 programs
- Compile the DB2LIB, DB2LIB2, and the GenArc programs
- Bind DB2LIB and DB2LIB2 into packages
- Bind the DB2LIB and DB2LIB2 packages into a plan
- Link-edit GenArc into an executable program, placing it in LINKLIB.

Depending on your DB2 system file names and C compiler file names, you will probably have to modify these procedures to meet your needs. The COMPDB2 job should finish with return codes of 00 or 04 in all steps.

Modifying the FSISYS INI File for Host Archive

Just as the FSIUSER and FSISYS initialization files provide the GenTrn, GenData, and GenPrint programs with configuration information, they also provide configuration information to the GenArc program.

The following control group and option settings must exist in either the FSIUSER or FSISYS members of DEFLIB. The FSISYS file shipped with the base product contains the values shown below. You only need to change these options if you have modified the names of the DB2 objects, such as the database name, table name, and so on.

ArcRet control group

The ArcRet control group contains information about the file (or table) names used for archiving information. The values for table names in this group cannot exceed eight characters. To specify a table name with more than eight characters use the DB2_FileConvert control group to map the eight character name you enter here to a longer (up to 18 characters) name.

```
< ArcRet >
AppIdxDfd = APPIDX
AppIdx    = APPIDX
CARFile   = ARCHIVE
CATALOG   = CATALOG
RestartTable= RESTART
```

Archival control group

The Archival control group indicates whether archived information is stored via a DBMS. For DAP archive to DB2, specify:

```
< Archival >
ArchiveMem = Yes
```

DBHandler:DB2 control group

The DBHandler:DB2 control group tells the GenArc program that DB2 will be used to access the tables used during the GenArc process.

The Database option identifies the name of the DB2 database that contains the tables used by the GenArc program. Database names can consist of up to eight characters.

The CreateTable option indicates whether the GenArc program can create the tables it needs or if the tables must exist before it begins.

Enter Description

No	Do not allow the GenArc program to create the DB2 tables.
Yes	Allow the GenArc program to create the DB2 tables.

The CreateIndex option indicates whether indexes are to be created for the APPIDX DB2 table if the APPIDX (DFD) member of DEFLIB contains keys information.

Enter Description

No	Do not allow the GenArc program to create the DB2 indexes.
Yes	Allow the GenArc program to create the DB2 indexes.

```
< DBHandler:DB2 >
  CreateTable = No
  CreateIndex = No
```

DB2_FileConvert control group

The DB2_FileConvert control group lets you use DB2 table names that are longer than eight characters. The option on the left (APPIDX) is the value associated with the corresponding option in the [ArcRet control group on page 27](#). The value on the right (DAP110_APP_R1) is the name of the DB2 table as it exists in the DB2 subsystem. If your table names are not longer than eight characters, you do not need to make any entries in the DB2_FileConvert control group.

```
< DB2_FileConvert >
  APPIDX      = DAP110_APP_R1
  Archive     = DAP110_ARC_R1
  Catalog     = DAP110_CAT_R1
  Restart     = DAP110_RES_R1
```

DBTable control group

Use the DefaultTag option to specify the default tag for ODBC and DB2. This tag is used by the ORDER BY clause in the SQL database to sort records.

```
< DBTable:MYTABLE >
  DefaultTag =
```

For the DefaultTag option, enter the name of the key from the DFD file. Keep in mind this only works with ODBC and DB2. It does not work with xBase files.

Trigger2Archive control group

The Trigger2Archive control group designates the fields in the NEWTRN DAT file (right side) that are copied to the columns in the APPIDX table (left side) during the GenArc process. Remember that the APPIDX table columns are defined by the APPIDX member of DEFLIB and that this APPIDX member is in a DFD format. For the sample resources, the Trigger2Archive control group might be set up as follows:

```
< Trigger2Archive >
```

```
Company      = Company  
Lob          = Lob  
PolicyNum    = PolicyNum  
RunDate      = RunDate
```

Changing the JCL for Host Archive

Examine the GenArcDX member of JCLLIB. Make sure the JCL is appropriate for your site and that the dataset names correspond to the names you have used for your files. Additionally, make sure the DB2 subsystem name and other DB2 object names, such as the plan names, are correct.

ARCHIVING TO DOCUMANAGE

Configuring the system to archive to Documanager involves performing these tasks:

- Setting up Documanager (see the Documanager manuals for more information)
- Modifying the FSISYS file (FSISYSM member in DEFLIB)
- Adding IP addresses and system/domain names of the Documanager machine to the TCPIP host data sets
- Making JCL changes

Modifying the FSISYS File

NOTE All references of the FSISYS file in this topic refer to FSISYSM member in DEFLIB.

In the ArcRet control group, the FSISYS file includes the values shown here:

```
< ArcRet >
  AppIdxDfd      = AppIdx
  AppIdx         = APPIDX
  CARFile        = ARCHIVE
  CARFileDFD     = CARFILE
  CARPath        =
; Catalog        = CATALOG
; RestartTable   = RESTART
```

For archiving to Documanager, you need to set the CARFileDFD option to CARFILE (which resides in DEFLIB) and comment out both the Catalog and RestartTable options.

The Archival control group in the FSISYS file should have these values:

```
< Archival >
  ArchiveMem      = Yes
  UseRestartTable = No
```

For archiving to Documanager, set the UseRestartTable option to No.

In the following DBTable control groups, the FSISYS file should have these values:

```
< DBTable:APPIDX >
  DBHandler       = DMIA
< DBTable:ARCHIVE >
  DBHandler       = DMIA
```

NOTE You must use *DMIA* as a DBHandler for archiving to Documanager.

The DBHandler:DMIA control group tells the GenArc program that DMIA is used to access Documanage during the GenArc process. Your FSISYS file includes these values:

```
< DBHandler:DMIA >
  RPCHost      = (IP address of the Documanage machine)
  Cabinet      = RPEX1OS390
  Domain       = (system/domain name)
  UserID       = (Documanage user ID)
  Password     = (Documanage password)
; Debug       = Max
```

You will need to add the IP address, system/domain name, user ID, and password to the appropriate INI options.

NOTE For whichever cabinet name is being set in the DBHandler:DMIA control group, you must include that name in the DMIA control group name. RPEX1OS390 is the cabinet name in this case. So, the DMIA control group name would be *DMIA:RPEX1OS390*

For the DMIA:RPEX1OS390 control group, the FSISYS file includes these values:

```
< DMIA:RPEX1OS390 >
  FileType     = DPA
  FolderBy     = company,lob,policynum
  NameDocBy    = arckey
  StackedDPA   = Yes
```

Adding IP Addresses and System/Domain Names

To add the IP addresses and system/domain names of the Documanage machine to the TCPIP host datasets, set the StackedDPA option to No when archiving to Documanage.

You will need to add the IP address and system/domain name of your Documanage machine to the local TCPIP host data sets on your OS/390. The steps below are examples you could follow to accomplish this task.

NOTE You first may need to consult your OS/390 system programmer on how to get this task done as the steps you need to do may differ from below.

- 1 Add an entry with your IP and system/domain name to hlq.HOSTS.LOCAL.
- 2 Execute a TSO MAKESITE command to generate hlq.HOSTS.SITEINFO and hlq.HOSTS.ADDRINFO datasets from your hlq.HOSTS.LOCAL.

Changing the JCL

Examine the GenArcX member of JCLLIB. Make sure the JCL is appropriate for your site and that the dataset names correspond to the names you have used for your files.

ARCHIVING USING STACKED DPA

Configuring the system to use stacked DPA to archive to a CARDATA file involves performing these tasks:

- Checking the FSISYS file (FSISYSM member of DEFLIB)
- Making JCL changes

For the DBHandler:DMIA control group below, you do not need to add IP address, system/domain name, user ID, or password to run GenArc with stacked DPA.

```
< DBHandler:DMIA >
  RPCHost      = (IP address of the Documange machine)
  Cabinet      = RPEX1OS390
  Domain       = (system/domain name)
  UserID       = (Documange user ID)
  Password     = (Documange password)
; Debug       = Max
```

In the DMIA:RPEX1OS390 control group, make sure the StackedDPA option is set to Yes.

```
< DMIA:RPEX1OS390 >
  FileType     = DPA
  FolderBy     = company,lob,policynum
  NameDocBy    = arckey
  StackedDPA   = Yes
```

NOTE All other INI options for archiving to Documange are the same for stacked DPA.

Changing the JCL

Examine the GenArcX member of JCLLIB. Make sure the JCL is appropriate for your site and that the data set names correspond to the names you have used for your files.

CHECKING YOUR INSTALLATION

To make sure the base system has been installed correctly you should:

- Check the INI files (FSIUSER and FSISYS).
- Check the JCL for the DAPRUN and DAP procedure (or GenTrnX, GenDataX, GenPrtX).
- Run DAPRUN (or GenTrnX, GenDataX, GenPrtX).
- Run archive (GenArcDX or GenArcX). This step is optional.
- Print the output and verify that it appears correct.

NOTE Two sets of resources are included on the installation tape. The RPEX1 resources are designed for the insurance market and the UTEX1 resources are designed for the utility market. You can verify your installation by running the system using one or both of these sets of resources. When you run the DAPRUN job later in this guide, you will need to specify as part of the jobcard, which set of resources the DAPRUN job is using.

You indicate the resource set by specifying the `SET RES=xxxxx` statement, where `xxxxx` is either `RPEX1` or `UTEX1`. By default, the JOBCARD is set up to use the RPEX1 resources.

CHECKING YOUR FSIUSER AND FSISYS SETTINGS

You loaded the FSIUSER and FSISYS members from the tape into the DEFLIB PDS during installation. These two files have similar control groups and options but different purposes. Generally, the FSIUSER member contains customer-specific initialization options while the FSISYS member contains general system-wide options.

Among other things, the FSIUSER and FSISYS members contain the names of various input and output files. On the PC, these file names include the path and file name. On MVS, these names consist of DD statement names or PDS member names or both.

The FSIUSER and FSISYS you uploaded as part of the sample resources are in a format you can run on MVS. To verify these values, consider the following but note that this list of options is not complete.

The relevant value in the FSIUSER INI file is...

```
< Environment >
  FSISYSINI = DD:FSISYS
```

The relevant values in FSISYS that indicate the DD names of various libraries, the PDS members, or are required for the system to run are:

```
< BATCH1 >
  Printer = Printer1
< BATCH2 >
  Printer = Printer2
< BATCH3 >
  Printer = Printer3
< Manual >
  Printer = Pmanual
< Error >
```

```
Printer = Perror
< CONFIG:Batch Processing >
  CompLib      = DD:COMPLIB()
  DDTLib       = DD:DDTLIB()
  DefLib       = DD:DEFLIB()
  FormDef      = FORM
  FormLib      = DD:FAPLIB()
  XrfFile      = REL95SM
< Control >
  ImageExt     = .FAP
  LogoExt      = .LOG
  XrfExt       = .FXR
< Data >
  DataPath     =
  ExtrFile     = DD:EXTRACT
  DbLogFile    = DD:DBLOGFILE
  ErrFile      = DD:ERRFILE
  LogFile      = DD:LOGFILE
  MsgFile      = DD:MSGFILE
  NAFile       = DD:NAFILE
  NewTrn       = DD:NEWTRN
  PolFile      = DD:POLFILE
  TrnFile      = DD:TRNFILE
  WIPBatch     = DD:MANUAL
  Manual       = DD:MANUAL
  TrnDfdFile   = TrnDfdFl
  AfgJobFile   = DD:AFGJOB
  RcbDfdFile   = RcbDfdFl
  SetRcpTb     = SetRcpTb
  TextTbl      = TextTbl
  TblFile      = TblFile
< Print_Batches >
  Batch1       = DD:BATCH1
  Batch2       = DD:BATCH2
  Batch3       = DD:BATCH3
  Manual       = DD:MANUAL
  Error        = DD:ERROR
< Printer >
  PrtType      = AFP
  ;PrtType     = XER
< Printer1 >
  PORT         = DD:PRTBAT1
< Printer2 >
  PORT         = DD:PRTBAT2
< Printer3 >
  PORT         = DD:PRTBAT3
< PManual >
  PORT         = DD:PMANUAL
< PError >
  PORT         = DD:PERROR
< PrinterInfo >
  Printer      = Printer1
  Printer      = Printer2
  Printer      = Printer3
  Printer      = PManual
```

```

Printer          = PError
< PrtType:XER >
  ColorCharts     = Yes
  CompileInstream = No
  DownLoadFonts   = No, Disabled
  InitFunc        = XERInit
  Module          = XERW32
  OutputFunc      = XEROutput
  OutMetFunc      = XEROutMet
  PageNumbers     = Yes
  PrintFunc       = XERPrint
  PrinterInk      = Blue
  TermFunc        = XERTerm
  SendColor       = Yes
  DJDEIden        = A'@@@DJDE'
  DJDEOffset      = 0
  DJDESkip        = 8
  Environment     = MVS
  OutMode         = JES2
  ImageOpt        = Yes
  JDEName         = DFLT
  JDLData         = 0, 255
  JDLHost         = IBMONL
  JDLName         = DEFAULT
  JDLCode         = NONE
  JDLRStack       = 0, 10, EQ, X'13131313131313131313'
  JDLRPage        = 0, 5, EQ, X'FFFF26FFFF'Printer control group
< PrtType:AFP >
  SendOverlays    = Yes
< TRN_File >
  MaxExtRecLen     = 119 (for RPEX1 only)
  MaxExtRecLen     = 450 (for UTEX1 only)
  BinaryExt        = N

```

CHECKING THE JCL

As stated in the topic [Configuring Your Printer Type on page 14](#), when you run the system with the sample resources you can run each step as a separate job (GenTrnX, GenDataX, GenPrintX) or together in sequence using the DAPRUN job which invokes the DAP procedure in PROCLIB.

For the verification test, we recommend you use the DAPRUN job. If you want to use the DAPRUN job and DAP procedure, inspect and modify those files. If you want to run GenTrnX, GenDataX, and GenPrtX separately, inspect and modify those files.

If you plan to use the DAPRUN job...

- 1 Check the DAPRUN job in JCLLIB. This job invokes the DAP procedure. Make sure DAPRUN has a valid jobcard and that the PROCLIB name is correct.
- 2 Check the DAP procedure in PROCLIB. This procedure contains these steps:

	Description
GENTRND	deletes files in preparation for the GenTrn program
GENTRN	runs the GenTrn program
GENDATAD	deletes files in preparation for the GenData program
GENDATA	runs the GenData program
GENPRTD	deletes files in preparation for the GenPrint program
GENPRT	runs the GenPrint program

- 3 Check the JCL statements in the DAP procedure and change the dataset names to match the names you chose during installation. Also, change any UNIT names to conform to your site's standards.

Remember that the DD names for various input files (DEFLIB, FAPLIB, TRNFILE, and so on) and output files (BATCH1, NAFILE, and so on) need to correspond to the values that point to these files in the FSISYS file.

You should not need to modify the actual DD names for these files either in the DAP procedure nor in the FSISYS member because these files have been set up to run the verification test as is.

Running GenTrnX, GenDataX, and GenPrtX separately

Look at the GenTrnX, GenDataX and GenPrtX jobs in JCLLIB and modify any jobcards, dataset names, unit names, and so on, so these jobs will run on your system.

Remember that the DD names for various input files (DEFLIB, FAPLIB, TRNFILE, and so on) and output files (BATCH1, NAFILE, and so on) should correspond to the values that point to these files in the FSISYS member. You should not need to modify the actual DD names for these files either in the JCL jobs or in the FSISYS member because these files are set up to run the verification test as is.

RUNNING DAPRUN (OR GENTRN, GENDATA, AND GENPRTX)

Submit the job DAPRUN. DAPRUN should receive return code `03` or `04` in each step. The output for DAPRUN should include logging messages, which are sent to SYSPRINT. Depending on whether you are using the RPEX1 or UTEX1 sample resources, the messages will differ. Examples of both are shown below and on the following pages.

NOTE If you are running GenTrnX, GenDataX, and GenPrtX separately you should submit each job in sequence and check the return codes.

Running DAPRUNC to Improve Performance

The DAPRUNC job invokes the DAPC procedure in PROCLIB. The DAPC procedure contains JCL that assumes the GenTrn and GenPrint steps have been combined into the GenData step. The GenTrn and GenPrint steps are run as rules within the GenData step. This is called *single-step processing* and can improve system performance. To use single-step processing, submit DAPRUNC. It should receive a return code `03` or `04` in both steps.

NOTE When using single-step processing, only the SYSOUT messages for GenData are displayed.

Logging messages when running RPEX1

```

--- GenTrn ---
Filtering extract file...
Building transaction file...
==> Processing: TransactionId-GroupName1-GroupName2-GroupName3-
TransactionType
==> Processing: 1234567-SAMPCO-LB1--T1
==> Processing: 2234567-SAMPCO-LB1--T1
==> Processing: 5SAMPCO-SAMPCO-LB2--T1
==> Processing: 6SAMPCO-SAMPCO-LB2--T1
==> Processing: 7SAMPCO-SAMPCO-LB2--T1
==> Processing: 8SAMPCO-SAMPCO-LB2--T1
==> Processing: 9SAMPCO-SAMPCO-LB2--T1
==> Processing: 4234567-FSI-CPP--T1
==> Processing: 5234567-FSI-GL--T1
==> Warning count:    0
==> Error count:      0
Elapsed Time: 3 seconds
--- GenTrn Completed ---

--- GenData ---
==> Processing: TransactionId-GroupName1-GroupName2-GroupName3-
TransactionType
==> Processing: 1234567-SAMPCO-LB1--T1
==> Processing: 2234567-SAMPCO-LB1--T1
==> Processing: 5SAMPCO-SAMPCO-LB2--T1
==> Processing: 6SAMPCO-SAMPCO-LB2--T1
==> Processing: 7SAMPCO-SAMPCO-LB2--T1
==> Processing: 8SAMPCO-SAMPCO-LB2--T1

```

```
==> Processing: 9SAMPCO-SAMPCO-LB2--T1
==> Processing: 4234567-FSI-CPP--T1
==> Processing: 5234567-FSI-GL--T1
==> Warning count:    0
==> Error count:      0
Elapsed Time: 44 seconds
--- GenData Completed ---

--- GenPrint ---
Processing Batch:<BATCH1> File:<DD:BATCH1> Port:<DD:XERBAT1>
==> Processing: TransactionId-GroupName1-GroupName2-GroupName3-
TransactionType
==> Processing: 1234567-SAMPCO-LB1--T1
==> Processing: 2234567-SAMPCO-LB1--T1
==> Processing: 5SAMPCO-SAMPCO-LB2--T1
==> Processing: 6SAMPCO-SAMPCO-LB2--T1
==> Processing: 7SAMPCO-SAMPCO-LB2--T1
==> Processing: 8SAMPCO-SAMPCO-LB2--T1
==> Processing: 9SAMPCO-SAMPCO-LB2--T1
==> Processing: 4234567-FSI-CPP--T1
==> Processing: 5234567-FSI-GL--T1
9 records processed in BATCH1 batch.
Processing Batch:<BATCH2> File:<DD:BATCH2> Port:<DD:XERBAT2>
==> Processing: 1234567-SAMPCO-LB1--T1
==> Processing: 2234567-SAMPCO-LB1--T1
==> Processing: 5SAMPCO-SAMPCO-LB2--T1
==> Processing: 6SAMPCO-SAMPCO-LB2--T1
==> Processing: 7SAMPCO-SAMPCO-LB2--T1
==> Processing: 8SAMPCO-SAMPCO-LB2--T1
==> Processing: 9SAMPCO-SAMPCO-LB2--T1
==> Processing: 4234567-FSI-CPP--T1
==> Processing: 5234567-FSI-GL--T1
9 records processed in BATCH2 batch.
Processing Batch:<BATCH3> File:<DD:BATCH3> Port:<DD:XERBAT3>
==> Processing: 1234567-SAMPCO-LB1--T1
==> Processing: 2234567-SAMPCO-LB1--T1
==> Processing: 5SAMPCO-SAMPCO-LB2--T1
==> Processing: 6SAMPCO-SAMPCO-LB2--T1
==> Processing: 7SAMPCO-SAMPCO-LB2--T1
==> Processing: 8SAMPCO-SAMPCO-LB2--T1
==> Processing: 9SAMPCO-SAMPCO-LB2--T1
7 records processed in BATCH3 batch.
Processing Batch:<ERROR> File:<DD:ERROR> Port:<DD:PERROR>
Processing Batch:<MANUAL> File:<DD:MANUAL> Port:<DD:PMANUAL>
==> Warning count:    3
==> Error count:      0
Elapsed Time: 49 seconds
--- GenPrint Completed ---
```

Logging messages when
running UTEX1

```

--- GenTrn ---
Filtering extract file...
Building transaction file...
==> Processing: TransactionId-GroupName1-GroupName2-GroupName3-
TransactionType
==> Processing: 2222-3445-UTIL-BILL--
==> Processing: 1234-5555-UTIL-BILL--
==> Processing: 3456-7777-UTIL-BILL--
==> Processing: 3999-9999-UTIL-BILL--
==> Processing: 4444-1234-UTIL-BILL--
==> Processing: 4999-R777-UTIL-BILL--
==> Processing: 5111-8888-UTIL-BILL--
==> Processing: 5555-7010-UTIL-BILL--
==> Processing: 6750-R758-UTIL-BILL--
==> Processing: 7000-7000-UTIL-BILL--
==> Processing: 2428-1927-UTIL-BILL--
==> Processing: 8234-5555-UTIL-BILL--
==> Warning count:    0
==> Error count:      0
Elapsed Time: 3 seconds
--- GenTrn Completed ---

--- GenData ---
==> Processing: TransactionId-GroupName1-GroupName2-GroupName3-
TransactionType
==> Processing: 2222-3445-UTIL-BILL--
==> Processing: 1234-5555-UTIL-BILL--
==> Processing: 3456-7777-UTIL-BILL--
==> Processing: 3999-9999-UTIL-BILL--
==> Processing: 4444-1234-UTIL-BILL--
==> Processing: 4999-R777-UTIL-BILL--
==> Processing: 5111-8888-UTIL-BILL--
==> Processing: 5555-7010-UTIL-BILL--
==> Processing: 6750-R758-UTIL-BILL--
==> Processing: 7000-7000-UTIL-BILL--
==> Processing: 2428-1927-UTIL-BILL--
==> Processing: 8234-5555-UTIL-BILL--
==> Warning count:    0
==> Error count:      0
Elapsed Time: 148 seconds
--- GenData Completed ---

--- GenPrint ---
Processing Batch:<BATCH1> File:<DD:BATCH1> Port:<DD:AFPBAT1>
==> Processing: TransactionId-GroupName1-GroupName2-GroupName3-
TransactionType
==> Processing: 2222-3445-UTIL-BILL--
==> Processing: 1234-5555-UTIL-BILL--
==> Processing: 3456-7777-UTIL-BILL--
==> Processing: 3999-9999-UTIL-BILL--
==> Processing: 4444-1234-UTIL-BILL--
==> Processing: 4999-R777-UTIL-BILL--
==> Processing: 5111-8888-UTIL-BILL--
==> Processing: 5555-7010-UTIL-BILL--
==> Processing: 6750-R758-UTIL-BILL--
==> Processing: 7000-7000-UTIL-BILL--

```

```
==> Processing: 2428-1927-UTIL-BILL--
==> Processing: 8234-5555-UTIL-BILL--
12 records processed in BATCH1 batch.
Processing Batch:<BATCH2> File:<DD:BATCH2> Port:<DD:AFPBAT2>
Processing Batch:<BATCH3> File:<DD:BATCH3> Port:<DD:AFPBAT3>
Processing Batch:<ERROR> File:<DD:ERROR> Port:<DD:PEERROR>
Processing Batch:<MANUAL> File:<DD:MANUAL> Port:<DD:PMANUAL>
==> Warning count:    5
==> Error count:      0
Elapsed Time: 75 seconds
--- GenPrint Completed ---
```

RUNNING GENARCDX

This task is optional.

The JCLLIB member GenArcDX (GenArcX prior to version 10.3) executes the GenArc program in two steps. The first step, called *GenArcD*, executes the IEFBR14 program and contains DD statements that delete various files which will be created in the following GenArc step. The second step, called *GenArc*, executes the TSO Terminal Monitor Program and starts the DB2 Command Processor program (DSN) which runs the GenArc program and DB2 plan.

At the bottom of the GenArcDX job, the SYSTSIN DD statement has some in-stream control cards, one of which indicates the LINKLIB you loaded from tape. Change &HLQ..LINKLIB to match the name you gave the LINKLIB you loaded from the tape. Also, you'll need to change the subsystem name from *TDB1* to match your DB2 subsystem name.

```
//SYSTSIN DD *
  DSN SYSTEM(TDB1)
  RUN PROGRAM(GENARC) PLAN(DAP110) -
    LIB('&HLQ..LINKLIB')
  END
```

To test, submit the GenArcDX job. The GenArcDX job should receive a return code of *00* or *04*. The output for the GenArcDX job should include logging messages (sent to SYSPRINT) similar to the following messages.

RPEX1 messages

```
--- GenArc ---
==> Processing: TransactionId-GroupName1-GroupName2-GroupName3-
TransactionType
==> Processing: 1234567-SAMPCO-LB1--T1
==> Processing: 2234567-SAMPCO-LB1--T1
==> Processing: 5SAMPCO-SAMPCO-LB2--T1
==> Processing: 6SAMPCO-SAMPCO-LB2--T1
==> Processing: 7SAMPCO-SAMPCO-LB2--T1
==> Processing: 8SAMPCO-SAMPCO-LB2--T1
==> Processing: 9SAMPCO-SAMPCO-LB2--T1
==> Processing: 4234567-FSI-CPP--T1
==> Processing: 5234567-FSI-GL--T1

==> Transactions Read      :      9
==> Transactions Archived :      9
==> Transactions In Error :      0
==> Transactions Rolled Back:      0

==> Warning count:      1
==> Error count:      0

Elapsed Time: 44 seconds
--- GenArc Completed ---
```

UTEX1 messages

```
--- GenArc ---

==> Processing: TransactionId-GroupName1-GroupName2-GroupName3-
TransactionType
==> Processing: 2222-3445-UTIL-BILL--
==> Processing: 1234-5555-UTIL-BILL--
==> Processing: 3456-7777-UTIL-BILL--
==> Processing: 3999-9999-UTIL-BILL--
==> Processing: 4444-1234-UTIL-BILL--
==> Processing: 4999-R777-UTIL-BILL--
==> Processing: 5111-8888-UTIL-BILL--
==> Processing: 5555-7010-UTIL-BILL--
==> Processing: 6750-R758-UTIL-BILL--
==> Processing: 7000-7000-UTIL-BILL--
==> Processing: 2428-1927-UTIL-BILL--
==> Processing: 8234-5555-UTIL-BILL--

==> Transactions Read      :      12
==> Transactions Archived :      12
==> Transactions In Error :        0
==> Transactions Rolled Back:        0

==> Warning count:      0
==> Error count:        0

Elapsed Time: 34 seconds
--- GenArc Completed ---
```

RUNNING GENARCX

This task is optional.

The JCLLIB member GenArcX executes the GenArc program in two steps. The first step, called GenArcD, executes the IEFBR14 program and contains DD statements that delete various files which will be created in the following GenArc step. The second step, called GenArc, executes the GenArc program to archive to Documange or to a CARDATA file if you are using stacked DPA.

To test, submit the GenArcX job. The GenArcX job should receive a return code of 04. The output for the GenArcX job should include logging messages (sent to SYSPRINT) similar to the following messages if you are using stacked DPA.

RPEX1 messages

```
--- GenArc ---
==> Processing: TransactionId-GroupName1-GroupName2-GroupName3-
TransactionType
==> Processing: 1234567-SAMPCO-LB1--T1
==> Processing: 2234567-SAMPCO-LB1--T1
==> Processing: 5SAMPCO-SAMPCO-LB2--T1
==> Processing: 6SAMPCO-SAMPCO-LB2--T1
==> Processing: 7SAMPCO-SAMPCO-LB2--T1
==> Processing: 8SAMPCO-SAMPCO-LB2--T1
==> Processing: 9SAMPCO-SAMPCO-LB2--T1
==> Processing: 4234567-FSI-CPP--T1
==> Processing: 5234567-FSI-GL--T1
==> Transactions Read      :      9
==> Transactions Archived :      7
==> Transactions In Error :      0
==> Transactions Rolled Back:      0
==> Warning count:      1
==> Error count:      0
Elapsed Time: 27 seconds
--- GenArc Completed ---
```

UTEX1 messages

```
--- GenArc ---
==> Processing: TransactionId-GroupName1-GroupName2-GroupName3-
TransactionType
==> Processing: -UTIL-BILL--
==> Processing: -UTIL-BILL--
==> Processing: -UTIL-BILL--
==> Processing: -UTIL-BILL--
==> Processing: -UTIL-BILL--
==> Processing: -UTIL-BILL--
==> Processing: -UTIL-BILL--
==> Processing: -UTIL-BILL--
==> Processing: -UTIL-BILL--
==> Processing: -UTIL-BILL--
==> Processing: -UTIL-BILL--
==> Transactions Read      :     12
==> Transactions Archived :     12
==> Transactions In Error :      0
==> Transactions Rolled Back:      0
==> Warning count:      1
==> Error count:      0
Elapsed Time: 66 seconds
--- GenArc Completed ---
```

PRINTING THE OUTPUT

When you performed the tasks discussed in the topic [Running DAPRUN \(or GenTrnX, GenDataX, and GenPrtX\)](#) on page 41, DAPRUN should have produced three print-ready files. These files would be referenced as shown in the following table:

If you configured your system for... You will see these default DD names...

Xerox Metacode	PRTBAT1, PRTBAT2, and PRTBAT3
IBM AFP	PRTBAT1, PRTBAT2, and PRTBAT3
PostScript	PRTBAT1, PRTBAT2, and PRTBAT3
PDF	PDFLIB(BAT10000), PDFLIB(BAT20000), and PDFLIB(BAT30000)

Consult the appropriate topic below to send the output to the type of printer you are configured for.

Xerox Metacode printer

If you configured the system to create Xerox Metacode, use the GENERXER member of JCLLIB to ~~EBGENER~~ the print files to JES. Make any changes necessary in this member to match your MVS environment.

BARR-attached Metacode printer

If you have a Metacode printer attached to your local area network (LAN) and you are using the BARR interface, you must first ~~BARRWRAP~~ the output print stream, then send this print stream to the printer's spool area.

To BARRWRAP the output print stream, run the BARRWRPX utility. You'll find an example of this utility in JCLLIB. Then download the resulting file to your LAN. Be sure to download without carriage returns or line feeds (CRLF) and without ASCII translation. Once the file is downloaded, copy it to the BARR SPOOL area.

IBM AFP printer

If you have configured the system to create an IBM AFP data stream, use the GENERAFP member of JCLLIB to ~~EBGENER~~ the print files to JES. Make any changes necessary in this member to match your MVS environment.

OPTIMIZING PERFORMANCE

This topic will help you configure your system for optimum performance. To gather the following recommendations, we first created benchmarks on a test system. Then, by changing different parameters of that system, we measured performance gains or losses. Here are some of the terms we used during this exercise:

CPU TIME. The amount of time that a program, such as GenTrn, GenData, or GenPrint, uses the CPU. In a multi-tasking operating system like MVS, there are many tasks competing for use of the CPU.

EXCB. Execute Channel Programs. We have used these counts as basic measurements of I/O activity.

WALLCLOCKTIME. The elapsed time, as measured from the time a program begins to the time that the program ends. ~~This~~ *all clock time* can vary significantly from one run to another.

BATCHWINDOW Most installations have specific times of the day or night when large batch processes, like this system, are scheduled to run. The time frame in which these processes run is sometimes referred to as ~~the~~ *batch window*. A batch window is measured in *wall clock time*, such as from 10:00 pm to 5:00 am. Your system installation should run fast enough to complete its processing within ~~the~~ *batch window*.

Most, but not all, of the following recommendations are the result of many tests and subsequent improvements designed for a hypothetical user. The characteristics of Documaker Server implemented for this hypothetical user are as follows:

- Extract file with large record length (approximately 25,000 bytes/record).
- Form sets composed with large number of individual images.
- Large number of different recipients (approximately 300).
- Moderate number of transactions (approximately 4,000)

COMPILEOPTIONS

If your license included source code, use the following information when compiling your system for best results.

OPTIMIZE Use the highest level of C optimization by passing the following option to the compiler:

Use	If you are using
-----	------------------

OPT(1)	C/C++ For MVS/ESA, 1 is the highest level
--------	---

OPT(2)	either of the other C/370 compilers
--------	-------------------------------------

NOVALIDATEHANDLE Specify NOVALIDATEHANDLE by passing the following option to the compiler:

DEF(NOVALIDATEHANDLE)

INLINE The INLINE option can produce performance gains but there are other activities required for it to work.

LANGUAGEENVIRONMENT(LE)OR C370 RUNTIMEOPTIONS

Use the following runtime options:

REPORT You can use the REPORT option to gather information on various values reached when you run a particular program, such as GenData or GenPrint. There is overhead involved in running with this option.

Do not use the REPORT option in production runs. Use the REPORT option during testing to report runtime values, then use these values to assess the need to run with additional runtime options such as HEAP.

HEAP Based on the values received from running with the REPORT option, you may choose to specify the HEAP run-time option, modifying the initial and incremental heap allocations as the values from REPORT would recommend. You would then want to remove the REPORT option.

If time permits, set HEAP values based on the values received from running with the REPORT option.

FILE TYPES AND CHARACTERISTICS

Block sizes Best results have been achieved by specifying large block sizes. In general, try to use half-track block size—that is, if you are creating files on a 3380-type device (which has a track size of 47,476 bytes), specify a block size around 23,500. A half-track block size for a 3390-type device (which has a track size of 56,664 bytes) would be around 28,000.

Specify half-track block sizes for all files.

Record formats *FB* stands for the *fixed block* record format and *FBS* stands for the Fixed Block Standard record format. The record format is specified on the JCL DD statement when a particular file is defined. The C function *fseek* appears to be able to seek more quickly to a location in a file if that file is defined with a record format of FBS instead of FB.

The *standard* keyword guarantees that all the blocks of a file will be of the stated block size length and that there will be no *short* blocks.

If you are choosing between record formats of FB and FBS for any of these files, choose FBS: EXTRACT, TRNFILE.DAT, NEWTRN.DAT, NAFILE.DAT, and POLFILE.DAT.

As for the size of record blocks, the larger the record length, the fewer records per block. For FBS, this results in additional EXCPs and slower performance.

Number of files As mentioned earlier, the total run-time, *wall clock time* of a job is increased as more files are created, opened, and closed. An additional 10 or 20 files may have no significant affect on the wall time but if your system has to create, open and close hundreds of files, the wall time will lengthen significantly.

Design your system with a minimal number of recipient batch and print files. For instance, try to keep the number of recipient files to less than 20 or so. See also the [AliasPrintBatches option on page 51](#)

Extract Files

The standard extract file is usually defined with a record format of fixed block standard (FBS) with a half-track block size.

Defining the extract file as a VSAM ESDS You may choose to define the extract file to be a VSAM Entry Sequenced Data Set (ESDS). See [Defining the Extract File as a VSAM ESDS on page 52](#) for more information.

DEFLIB

The standard DEFLIB is created as a Partitioned Data Set (PDS), or as a Partitioned Data Set Extended (PDS/E). The best performance results have been achieved with DEFLIB defined with a Variable Blocked (VB) record format.

Placing DDT files into a VSAM KSDS DEFLIB contains DDT files, DFD files, SETRCPTB, FORMDAT, various tables, and so on. You can move the DDT files into a VSAM KSDS by running the program PDS2VSAM. See [Moving DDT Files into a VSAM KSDS on page 64](#) for more information.

SETRCPTB

Moving the SETRCPTB member into a VSAM KSDS

You can move the set recipient table member, SETRCPTB, into a VSAM KSDS by running the RCP2VSAM utility. See [Moving SETRCPTB to a VSAM KSDS on page 66](#) for more information.

FAPLIB

Define the FAPLIB file as variable blocked (VB) with a half-track block size.

TRNFILE

Define the transaction trigger file (TRNFILE) file as fixed block standard (FBS) with a half-track block size.

NEWTRN

Define the updated transaction trigger file (NEWTRN) file as fixed block standard (FBS) with a half-track block size.

NAFILE

Define the NAFILE file as fixed block standard (FBS) with a half-track block size or use VSAM to create and read from the NAFILE as outlined below.

Creating a NAFILE as a VSAM file

The NAFILE can be created in the GenData step as a VSAM key sequenced data set (KSDS), then read in the GenPrint step. Actually, NAFILE and POLFILE function as a pair so that they must both be either VSAM or non-VSAM. See [Creating NAFILES and POLFILES as VSAM KSDSs on page 62](#) for the steps required to create the NAFILE/POLFILE files as VSAM files.

POLFILE

Define the POLFILE file as fixed block standard (FBS) with a half-track block size or use VSAM to create and read from the POLFILE as outlined in the section below.

Creating a POLFILE as a VSAM file

The POLFILE can be created in the GenData step as a VSAM key sequenced data set (KSDS), then read in the GenPrint step. See [Creating NAFILES and POLFILES as VSAM KSDSs on page 62](#) for the steps required to create the NAFILE.DAT and POLFILE.DAT files as VSAM files.

Recipient Batch Files

As mentioned earlier, the fewer recipient batch files you have the better the performance. Define the recipient batch files ~~fixed block standard~~ (FBS) with a half-track block size. A rule of thumb is to try to keep the number of recipient batch files under 20. If it is necessary to design the system as having many recipient batch files you may consolidate these logical groupings into a single ~~physical~~ recipient batch file. See [AliasPrintBatches option on page 56](#) for more information.

Pre-compiled Xerox Metacode Files (PMETLIB)

The standard PMETLIB is created as a PDS or PDS/E. If you are using a PDS or PDS/E, define PMETLIB with a record format of VB and with a half-track block size.

Placing PMETs into a VSAM KSDS

You can move the PMETs into a VSAM KSDS by running the program PDS2VSAM. See [Moving PMETs into a VSAM KSDS on page 58](#) for the steps to accomplish this task.

Print Files

Define the print files as Variable Blocked Machine (VBM) control codes, with a record length near 255 for Xerox or 8205 for AFP, and a half-track block size.

SETTING YOUR FSISYS INI OPTIONS

Caching Options

The following options attempt to minimize the repeated opening and closing of frequently used files (actually, PDS members in MVS) by retaining, or caching, file handles and file data. In many cases the defaults are sufficient but for specific cases, where many different Images are used, these caching values may be increased to improve performance.

Caching FAP files

In some cases, FAP files (image definitions) are loaded from the FORMLIB or FAPLIB PDS during GenData processing. The cache feature keeps frequently used FAP members available for re-use. Here is an example of how you set up the cache feature:

```
< Control >
CacheFAPFiles = 100
```

The default is 100. Accept the default value unless you are loading FAP files in GenData using the CompileInStream FSISYS INI option (set to Yes) and you are using more than 100 FAP files and/or logo files.

DDT files

Data definition table (DDT) members are loaded from the DDTLIB or DEFLIB PDS during GenData processing. The cache feature keeps frequently used DDT members available for re-use. Here is an example of how you set up the cache feature:

```
< Control >
RuleFilePool = 100
```

The default is 100. Accept the default value unless you are using more than 100 DDT files.

Using/Caching Xerox PMET files

MET files contain pre-compiled Xerox Metacode information produced by the FAP2MET utility. The GenPrint program loads MET members from the PMETLIB PDS. The cache option keeps frequently used MET members available for re-use. Here is an example of how you set up the cache feature:

```
< PrtType:XER >
  CacheFiles      = 100
  CompileInstream= No
```

The default is 100. Accept the default unless you are using pre-compiled FAP files you are using more than 100 FAP files and/or logo files.

Using AFP Overlays

For best performance, you should run the FAP2OVL utility, compiling FAPs into AFP overlays. Tell the system to use the overlays by specifying:

```
< PrtType:AFP >
  SendOverlays= Yes
```

LOGGING OPTIONS

LogTransactions option

The GenTrn, GenData, and GenPrint programs optionally place transaction information into a LOG file. In most situations, this information is not needed. The LogTransactions option is specified in:

```
< Control >
  LogTransactions = No
```

The default is Yes.

For optimum performance, specify **No**.

LogToConsole option

The GenTrn, GenData, and GenPrint programs optionally place transaction information in the SYSPRINT DD statement you specified in the job's JCL. In most situations, this information is not needed. The LogToConsole option is specified in:

```
< Control >
  LogToConsole = No
```

The default is Yes.

For optimum performance, specify **No**.

DEBUGOPTIONS

If_Rule control group

```
< If_Rule >
  Debug_if = No
```

The default is Yes.

For optimum performance, specify **No**.

RunMode control group

For these run options:

```
< RunMode >
  DownloadFAP = No
  LoadCordFAP = No
  CompiledFAP = Yes
```

The defaults are...

```
DownloadFAP = No
LoadCordFAP = No
CompiledFAP = Yes
```

Set the DownLoadFAP option to **No**, the LoadCordFAP option to **No**, and the CompileFAP option to **Yes** for the best performance.

OTHEROPTIONS

MaxRecsPerTransaction option

```
< ExtractFile >
  MaxRecsPerTransaction = nn
```

The default is zero (0) and there is no maximum. Be careful using this option. You might want to use this option if you *know* that each record in the extract file corresponds to a transaction.

AliasPrintBatches option

```
< ExtractFile >
  AliasPrintBatches = Yes
```

The default is No. Use the default.

DEFINING THE EXTRACT FILE AS VSAM ESDS

The extract file is generally defined as a Fixed Blocked Standard (FBS) file, with the record length depending on the particular implementation. There are circumstances when you might want to define the extract file as a VSAM Entry Sequenced Data Set (ESDS) instead of an FBS file.

Follow the steps below to move the extract file into a VSAM ESDS.

- 1 Execute the job in member EXTESDS in the JCLLIB dataset. Modify the member as appropriate for your site. The EXTESDS job first uses the IBM program IDCAMS to delete and re-define the VSAM extract file and then uses IDCAMS to copy the (non-VSAM) extract file to the VSAM extract file. Set the RecordSize parameter to the exact record length of the extract file. This job should receive a condition code of 00 in each step.
- 2 Since the name of the VSAM extract file will probably differ from the name of the non-VSAM extract file, change the DAP procedure in PROCLIB (or the GenTrnX and GenDataX JCLLIB members if you are using those jobs) to reference the new name of the extract file. There should be a reference to the extract file in both the GenTrn and GenData steps of the DAP procedure.
- 3 Modify the FSISYS INI file as follows:

```
< TRN_FILE >
MaxExtReclen= (the exact record size of the VSAM extract file. The
               maximum LRECL (Logical Record Length) for a dataset
               is 32760-32K minus 8 bytes)
BinaryExt=YES
```

The EXTESDS member of JCLLIB is shown below:

```
/* COPY JOBCARD HERE
/* * * * * *
/*          EXTESDS -
/*
/*          THIS JOB PERFORMS 2 STEPS :
/*
/*          1. DELETE / DEFINE NEW EXTRACT FILE AS VSAM ESDS.
/*          2. REPRO EXISTING SEQUENTIAL EXTRACT FILE INTO NEW
/*             ESDS EXTRACT FILE.
/*          3. &HLQ..&RES.. CANNOT BE USED INSTREAM, PLEASE
/*             REPLACE THE OCCURRENCES IN THE SYSIN DD
/*             STATEMENT WITH THE APPROPRIATE VALUES.
/*
/* * * * * *
//DEFINE EXEC PGM=IDCAMS,REGION=4M
//SYSPRINT DD SYSOUT=*
//SYSIN     DD *

      DEL      &HLQ..&RES..EXTRACT.ESDS

      DEFINE CLUSTER(NAME(&HLQ..&RES..EXTRACT.ESDS) +
                    CYL(25 5) +
                    NONINDEXED +
                    REUSE ) +
```

```
DATA (NAME (&HLQ..&RES..EXTRACT.ESDS.DATA) +  
      RECORDSIZE (??? ???) )  
  
IF LASTCC = 0 THEN SET MAXCC = 0  
  
/*  
/*  
//REPRO EXEC PGM=IDCAMS,REGION=4M  
//SYSPRINT DD SYSOUT=*  
//DD1      DD DSN=&HLQ..&RES..EXTRACT,DISP=SHR  
//DD2      DD DSN=&HLQ..&RES..EXTRACT.ESDS,DISP=SHR  
//SYSIN    DD *  
      REPRO INFILE (DD1) OUTFILE (DD2)
```

MOVING DDT FILES INTO VSAM KSDS

DDT files normally reside as members in the DDTLIB or DEFLIB PDS. The DEFLIB PDS also contains other types of members, such as SETRCPTB, FORMDAT, FXR, FSISYS and various tables. These other types of members should not be copied to the VSAM KSDS, but it probably won't hurt anything if they are copied—so step 1 is optional.

Follow the steps below to move the DDTs in DEFLIB to a VSAM KSDS.

- 1 You can use the ALLOCDDT job in JCLLIB to define a "DDTLIB PDS" (to contain only DDT files) and to copy the DDT files for the sample resource from DEFLIB into this new PDS. Once you have all the DDT files in a separate (non-VSAM) PDS, proceed to step 2.

If you are moving your own resources (not the sample resources), copy your DDT files into this DDTLIB.

- 2 Execute the PDS2VSDX job that resides in the JCLLIB PDS. The job consists of these steps:

The IDCAMS step deletes/re-defines the VSAM KSDS that the DDT files will be copied into.

The PDS2VSAM step reads the DDTLIB PDS created in step 1 and copies the DDT files to the VSAM KSDS defined in the IDCAMS step.

The PDS2VSDX job is shown below:

```

/* * COPY JOBCARD HERE
/* * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * *
/* * JOB PERFORMS 2 STEPS :
/* *
/* *      1. DELETES / RE-DEFINES DDTVSAM KSDS.
/* *      2. RUNS PDS2VSAM PROGRAM TO COPY DDT MEMBERS INTO
/* *          DDTVSAM
/* *
/* *      PRIOR TO RUNNING THIS JOB YOU SHOULD HAVE RUN
/* *      THE JOB IN THE "ALLOCDDT" MEMBER OF JCLLIB TO
/* *      ALLOCATE A NON-VSAM DDTLIB AND COPY THE DDT'S
/* *      FROM DEFLIB INTO IT.
/* *
/* * NOTE: THE &HLQ AND &RES WITHIN INSTREAM JCL WILL NOT
/* *          WORK. PLEASE REPLACE THE &HLQ AND &RES WITH
/* *          THEIR RESPECTIVE VALUES.
/* *
/* * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * *
/* *
//IDCAMS EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//SYSUDUMP DD SYSOUT=*
//SYSIN    DD *

        DELETE &HLQ..&RES..DDTVSAM CLUSTER

        DEFINE CLUSTER
              ( NAME(&HLQ..&RES..DDTVSAM)
              CYL(25 5)
              REUSE

```

```

        SHAREOPTIONS(2 3) )           -
DATA                                  -
    ( NAME(&HLQ..&RES..DDTVSAM.DATA) -
      RECORDSIZE(2048 2048)           -
      CONTROLINTERVALSIZE(4096)       -
      FREESPACE(0 10)                 -
      KEYS(16 0) )                     -
INDEX                                -
    ( NAME(&HLQ..&RES..DDTVSAM.INDEX) )

IF LASTCC = 00 THEN SET MAXCC = 00

/*
//PDS2VSAM EXEC PGM=PDS2VSAM
//STEPLIB DD DISP=SHR,DSN=&HLQ..LINKLIB
//          DD DISP=SHR,DSN=SYS1.SCEERUN
//SYSOUT DD SYSOUT=*
//SYSPRINT DD SYSOUT=*
//PDSIN   DD DSN=&HLQ..&RES..DDTLIB,DISP=SHR
//VSAMOUT DD DSN=&HLQ..&RES..DDTVSAM,DISP=SHR

```

3 Add this option to the FSISYS INI file:

```

< VSAM >
    DDTVSA = DD:DDTVSAM

```

The DD name you use to associate with the VSAM DDT file is up to you. This INI option tells the system to open and read the DDT file using VSAM.

4 Modify the JCL for the GenData step so the DDTVSA DD statement is in that step and references the VSAM KSDS created for the DDT files in step 2. If you are using the DAP procedure in PROCLIB, add the appropriate DD statement to that procedure.

If, you are executing each step as a separate job, add the DD statement to the GenDataX job.

MOVING JDTs INTO AVSAM KSDS

The Job Definition Table (JDT) normally resides as member AFGJOB in the DEFLIB PDS. You can move this member (and other JDT members if you have several) into its own VSAM KSDS, similar to the VSAM KSDS used for DDT files.

Follow these steps to move the JDT files in DEFLIB to a VSAM KSDS.

- 1 You can use the ALLOCJDT job in JCLLIB to define a JDTLIB PDS (to contain only JDT files) and to copy the JDT files for the sample resource from DEFLIB into this new PDS. Once you have all the JDTs in a separate (non-VSAM) PDS proceed to step 2.

If you are moving your own resources (not the sample resources), copy your JDT files into this JDTLIB.

- 2 Execute the PDS2VSJX job that resides in the JCLLIB PDS. The job consists of these steps:

The IDCAMS step deletes/re-defines the VSAM KSDS that the JDT files will be copied into.

The PDS2VSAM step reads the JDTLIB PDS created in step 1 and copies the JDTs to the VSAM KSDS defined in the IDCAMS step.

The PDS2VSJX job is shown below:

```
//* COPY JOBCARD HERE
/* * * * *
/* JOB PERFORMS 2 STEPS :
/*
/*      1. DELETES / RE-DEFINES JDTV SAM KSDS.
/*      2. RUNS PDS2VSAM PROGRAM TO COPY JDT MEMBERS INTO
/*          JDTV SAM
/*
/*      PRIOR TO RUNNING THIS JOB YOU SHOULD HAVE RUN THE
/*      JOB IN MEMBER "ALLOCJDT" OF JCLLIB TO ALLOCATE A
/*      NON-VSAM JDTLIB AND COPY THE JDT MEMBER(S) INTO IT.
/*
/* NOTE: THE &HLQ AND &RES WITHIN INSTREAM JCL WILL NOT
/*      WORK. PLEASE REPLACE THE &HLQ AND &RES WITH
/*      THEIR RESPECTIVE VALUES.
/*
/* * * * *
/*
/* IDCAMS EXEC PGM=IDCAMS
/*SYSPRINT DD SYSOUT=*
/*SYSUDUMP DD SYSOUT=*
/*SYSIN   DD *

DELETE &HLQ..&RES..JDTV SAM CLUSTER

DEFINE CLUSTER
( NAME(&HLQ..&RES..JDTV SAM)
CYL(25 5)
REUSE
SHAREOPTIONS(2 3) )
DATA
( NAME(&HLQ..&RES..JDTV SAM.DATA) -
```

```

        RECORDSIZE(2048 2048)          -
        CONTROLINTERVALSIZE(4096)      -
        FREESPACE(0 10)                 -
        KEYS(16 0)                      -
INDEX                                     -
      ( NAME(&HLQ..&RES..JDTVVSAM.INDEX) )

      IF LASTCC = 00 THEN SET MAXCC = 00

/*
//PDS2VSAM EXEC PGM=PDS2VSAM
//STEPLIB DD DISP=SHR,DSN=&HLQ..LINKLIB
//          DD DISP=SHR,DSN=SYS1.SCEERUN
//SYSOUT DD SYSOUT=*
//SYSPRINT DD SYSOUT=*
//PDSIN   DD DSN=&HLQ..&RES..JDTLIB,DISP=SHR
//VSAMOUT DD DSN=&HLQ..&RES..JDTVVSAM,DISP=SHR

```

3 Add this option to the FSISYS INI file:

```

< VSAM >
      JDTVVSAM = DD:JDTVVSAM

```

The DD name you associate with the VSAM JDT file is up to you. This INI option tells the system to open and read the JDT file using VSAM.

4 Modify the JCL for the GenData step so that the JDTVVSAM DD statement is in that step and references the VSAM KSDS that was created for the JDT files in step 2. If you are using the DAP procedure in PROCLIB add the appropriate DD statement to that procedure. If, on the other hand, you are executing each step as a separate job, add the DD statement to the GenDataX job.

MOVING PMETs INTO AVSAM KSDS

Pre-compiled Metacode files normally reside as members in the PMETLIB PDS. Follow the steps below to move the PMETs in PMETLIB to a VSAM KSDS.

- 1 Execute the following PDS2VSPX job, changing any necessary values to fit your environment. This job resides in the JCLLIB PDS and consists of these steps:

The IDCAMS step deletes/re-defines the VSAM KSDS the PMETs will be copied into.

The PDS2VSAM step reads the PMETLIB PDS and copies the PMET members to the VSAM KSDS defined in the previous step.

```

/* COPY JOBCARD HERE
/* * * * * *
/* JOB PERFORMS 2 STEPS :
/*
/*      1. DELETES / RE-DEFINES PMETVSAM KSDS.
/*      2. RUNS PDS2VSAM PROGRAM TO COPY PMET MEMBERS INTO
/*          PMETVSAM
/*      3. THE DEFAULT INPUT / OUTPUT DD NAMES ARE "PDSIN" AND
/*          "VSAMOUT", RESPECTIVELY. IF YOU WISH TO OVERRIDE
/*          THESE NAMES PASS THE EXECUTE PARAMETER :
/*          PARM=' / I=DD:INPUTDD /O=DD:OUTPUTDD'
/*
/* NOTE: THE &HLQ AND &RES WITHIN INSTREAM JCL WILL NOT
/*       WORK. PLEASE REPLACE THE &HLQ AND &RES WITH
/*       THEIR RESPECTIVE VALUES.
/*
/* * * * * *
/*
//IDCAMS EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//SYSUDUMP DD SYSOUT=*
//SYSIN    DD *

DELETE &HLQ..&RES..PMETVSAM CLUSTER

DEFINE CLUSTER                                -
( NAME(&HLQ..&RES..PMETVSAM)                -
  CYL(25 25)                                -
  REUSE                                      -
  SHAREOPTIONS(2 3) )                        -
DATA                                         -
( NAME(&HLQ..&RES..PMETVSAM,DATA)            -
  RECORDSIZE(2048 2048)                     -
  CONTROLINTERVALSIZE(4096)                 -
  FREESPACE(0 10)                           -
  KEYS(16 0) )                              -
INDEX                                       -
( NAME(&HLQ..&RES..PMETVSAM,INDEX) )

IF LASTCC = 00 THEN SET MAXCC = 00

/*
/*

```

```
//PDS2VSAM EXEC PGM=PDS2VSAM
//STEPLIB DD DISP=SHR,DSN=&HLQ..LINKLIB
//          DD DISP=SHR,DSN=SYS1.SCEERUN
//SYSOUT DD SYSOUT=*
//SYSPRINT DD SYSOUT=*
//PDSIN DD DSN=&HLQ..&RES..PMETLIB,DISP=SHR
//VSAMOUT DD DSN=&HLQ..&RES..PMETVSAM,DISP=SHR
```

2 Add this option to the FSISYS INI file:

```
< VSAM >
    METVSAM = DD:PMETVSAM
```

The DD name you associate with the VSAM PMET file is up to you. This INI option tells the system to open and read the PMET file using VSAM.

3 Modify the JCL for the GenPrint step so the PMETVSAM DD statement is in that step and it references the VSAM KSDS created for the PMETs in step 1. If you are using the DAP procedure in PROCLIB, add the appropriate DD statement to that procedure. If you are executing each step as a separate job, add the DD statement to the GenPrtX job.

MOVING SETRCPTB TO A VSAM KSDS

Follow the steps below to move the SETRCPTB member from DEFLIB to a VSAM KSDS.

NOTE Before you run this, please make sure that there are no blank lines at the end of your extract file. Blank lines can cause the following error:

```
DMI002: Error: Required INI definition omitted.
Cannot locate INI group <Key1Table> with value = defined.
```

The error appears when you run Documaker Server.

- 1 Execute the following RCP2VSMX job, changing any necessary values to fit your installation. This job resides in the JCLLIB PDS and consists of these steps:

The IDCAMS step deletes/re-defines the VSAM KSDS that the SETRCPTB will be copied into.

The RCP2VSAM step reads the DEFLIB PDS and copies the SETRCPTB member to the VSAM KSDS defined in the previous step.

```
/* COPY JOBCARD HERE
/* * * * * *
/* JOB PERFORMS 2 STEPS :
/*
/*      1. DELETES / RE-DEFINES SETRCPVS KSDS.
/*      2. RUNS RCP2VSAM PROGRAM TO COPY SETRCPTB TO
/*          SETRCPVS KSDS.
/*
/*      DEFAULT INPUT / OUTPUT DD NAMES ARE SETRCPTB AND
/*      SETRCPVS, RESPECTIVELY. YOU CAN OVERRIDE BY
/*      PASSING THE PARM : PARM='/ I=DD:DDIN /O=DD:DDOUT'
/*
/* NOTE: THE &HLQ AND &RES WITHIN INSTREAM JCL WILL NOT
/*      WORK. PLEASE REPLACE THE &HLQ AND &RES WITH
/*      THEIR RESPECTIVE VALUES.
/*
/* * * * * *
/*
/*DEFINE EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//SYSUDUMP DD SYSOUT=*
//SYSIN    DD *

DELETE &HLQ..&RES..SETRCPVS CLUSTER

DEFINE CLUSTER
( NAME(&HLQ..&RES..SETRCPVS)
  RECORDSIZE(500 1000)
  KEYS(152 0)
  REUSE
  SHAREOPTIONS(2 3) )
DATA
( NAME(&HLQ..&RES..SETRCPVS.DATA)
  CONTROLINTERVALSIZE(4096)
```

```

        RECORDS(1000 1000)          -
        FREESPACE(0 5) )           -
INDEX                                -
    ( NAME(&HLQ..&RES..SETRCPVS.INDEX) -
      RECORDS(100 50) )

    IF LASTCC = 00 THEN SET MAXCC = 00
/*
//RCP2VSAM EXEC PGM=RCP2VSAM
//STEPLIB DD DISP=SHR,DSN=&HLQ..LINKLIB
//          DD DISP=SHR,DSN=SYS1.SCEERUN
//SETRCPTB DD DISP=SHR,DSN=&HLQ..&RES..DEFLIB(SETRCPTB)
//SETRCPVS DD DISP=SHR,DSN=&HLQ..&RES..SETRCPVS
//SYSPRINT DD SYSOUT=*

```

2 Add this option to the FSISYS INI file:

```

< VSAM >
      VSAMRCPTB = DD:SETRCPVS

```

The DD name you associate with the VSAM SETRCPTB file is up to you. This INI option tells the system to open and read the SETRCPTB file using VSAM.

3 Modify the JCL for the GenData step so the SETRCPVS DD statement is in that step and it references the VSAM KSDS created for the SETRCPTB in step 1. If you are using the DAP procedure in PROCLIB, add the appropriate DD statement to that procedure. If you are executing each step as a separate job, add the DD statement to the GenDataX job.

CREATING NAFILE AND POLFILE AS VSAM KSDS

Normally, the NAFILE and POLFILE, which are created and written to during the GenData step, are defined as sequential files with a record format of FBS. You can, however, create them as VSAM KSDSs. The member, NAPOLVS, in JCLLIB provides the JCL for the additional IDCAMS job step you will need. This JCL is also listed below:

```
//NAPOLVS EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//SYSIN      DD *

DELETE &HLQ..&RES..GENDATA.NAFILE.KSDS PURGE
DELETE &HLQ..&RES..GENDATA.POLFILE.KSDS PURGE

DEFINE CLUSTER(NAME(&HLQ..&RES..GENDATA.NAFILE.KSDS) +
              CYL(5 1)      +
              KEY(4 0)      +
              REUSE )      +
        DATA(NAME(&HLQ..&RES..GENDATA.NAFILE.KSDS.DATA) +
              RECORDSIZE(2048 2048) )      +
        INDEX(NAME(&HLQ..&RES..GENDATA.NAFILE.KSDS.INDEX) )

DEFINE CLUSTER(NAME(&HLQ..&RES..GENDATA.POLFILE.KSDS) +
              CYL(5 1)      +
              KEY(4 0)      +
              REUSE )      +
        DATA(NAME(&HLQ..&RES..GENDATA.POLFILE.KSDS.DATA) +
              RECORDSIZE(2048 2048) )      +
        INDEX(NAME(&HLQ..&RES..GENDATA.POLFILE.KSDS.INDEX) )
```

To use VSAM for the NAFILE and POLFILE follow these steps:

- 1 Insert the NAPOLVS JCL step listed above into the GenData step prior to the GenData execution step, changing any file names or other values to conform to your environment.
- 2 Make sure the NAFILE and POLFILE file names in the GenData and GenPrint steps (GenDataX and GenPrtX if you are executing each step as a separate job) reflect the name of the VSAM files created in the IDCAMS step.
- 3 Add this option to the FSISYS INI file:

```
< VSAM >
VSAMNA = DD:NAFILE
```

The DD name you associate with the VSAM NAFILE file is up to you. This INI option tells the system to open and read both the NAFILE and the POLFILE using VSAM.

THELOADDAP JOB

The LOADDAP job in JCLLIB is listed below. This job creates the...

- system datasets
- RPEX1 and UTEX1 sample datasets (for the verification test)
- font datasets

In addition, the LOADDAP job reads from the install tape and populates these datasets.

```

/* COPY JOBCARD HERE ...
/* * * * * *
/*          LOADDAP -
/*
/*          COPY THE DAP BATCH SYSTEM FILES, RPEX1 AND UTEX1 RESOURCES
/*          FROM A 3480 CARTRIDGE TO AN MVS SYSTEM. FILES LOADED ARE :
/*
/*          1. JCLLIB
/*          2. PROCLIB
/*          3. SOURCE
/*          4. INCLIB
/*          5. OBJLIB
/*          6. LINKLIB
/*          7. DBRMLIB (DB2)
/*          8. DEFLIB (RPEX1)
/*          9. DDTLIB (RPEX1)
/*          10. FAPLIB (RPEX1)
/*          11. COMPLIB (RPEX1)
/*          12. EXTRACT (RPEX1)
/*          13. LOG2IMG (RPEX1 XEROX LOGO IMAGES)
/*          14. LOG2FNT (RPEX1 XEROX LOGO FONTS)
/*          15. PFRMLIB (RPEX1 XEROX FRM'S)
/*          16. DEFLIB (UTEX1)
/*          17. DDTLIB (UTEX1)
/*          18. FAPLIB (UTEX1)
/*          19. COMPLIB (UTEX1)
/*          20. EXTRACT (UTEX1)
/*          21. LOG2IMG (UTEX1 XEROX LOGO IMAGES)
/*          22. LOG2FNT (UTEX1 XEROX LOGO FONTS)
/*          23. AFP240 (AFP 240-DPI FONTS)
/*          24. AFP300 (AFP 300-DPI FONTS)
/*          25. XEROX (XEROX FONTS)
/*          26. XEROXR (XEROX ROTATED FONTS)
/*
/*
/*          => GO TO BOTTOM OF JOB AND CHANGE VOLSER & CUSTHLQ TO MATCH
YOUR
/*          => SITE REQUIREMENTS.
/*
/* * * * * *
/*
/*          IN-STREAM PROC FOLLOWS ...
/*
//LOAD      PROC TUNIT=,
//          TVOLSER=,
//          CUSTHLQ=,
//          DUNIT=,

```

```

//          PDSTYPE=
//*
//JCLLIB EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=&HLQ..JCLLIB,DISP=SHR,UNIT=&TUNIT,
//          VOL=(,RETAIN,,,SER=&TVOLSER),LABEL=(1,SL)
//SYSUT2 DD DSN=&CUSTHLQ..JCLLIB,DISP=(,CATLG),UNIT=&DUNIT,
//          SPACE=(CYL,(2,1,10)),DSNTYPE=&PDSTYPE
//SYSIN DD DUMMY
//*
//PROCLIB EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=&HLQ..PROCLIB,DISP=SHR,UNIT=&TUNIT,
//          VOL=(,RETAIN,,,SER=&TVOLSER),LABEL=(2,SL)
//SYSUT2 DD DSN=&CUSTHLQ..PROCLIB,DISP=(,CATLG),UNIT=&DUNIT,
//          SPACE=(CYL,(2,1,10)),DSNTYPE=&PDSTYPE
//SYSIN DD DUMMY
//*
//SOURCE EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=&HLQ..SOURCE,DISP=SHR,UNIT=&TUNIT,
//          VOL=(,RETAIN,,,SER=&TVOLSER),LABEL=(3,SL)
//SYSUT2 DD DSN=&CUSTHLQ..SOURCE,DISP=(,CATLG),UNIT=&DUNIT,
//          SPACE=(CYL,(8,10,100)),DSNTYPE=&PDSTYPE
//SYSIN DD DUMMY
//*
//INCLIB EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=&HLQ..INCLIB,DISP=SHR,UNIT=&TUNIT,
//          VOL=(,RETAIN,,,SER=&TVOLSER),LABEL=(4,SL)
//SYSUT2 DD DSN=&CUSTHLQ..INCLIB,DISP=(,CATLG),UNIT=&DUNIT,
//          SPACE=(CYL,(4,1,50)),DSNTYPE=&PDSTYPE
//SYSIN DD DUMMY
//*
//OBJLIB EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=&HLQ..OBJLIB,DISP=SHR,UNIT=&TUNIT,
//          VOL=(,RETAIN,,,SER=&TVOLSER),LABEL=(5,SL)
//SYSUT2 DD DSN=&CUSTHLQ..OBJLIB,DISP=(,CATLG),UNIT=&DUNIT,
//          SPACE=(CYL,(35,5,75)),DSNTYPE=&PDSTYPE
//SYSIN DD DUMMY
//*
/* DEFINE LINKLIB AS PDS (NOT PDSE) BY DEFAULT
/*
//LINKLIB EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=&HLQ..LINKLIB,DISP=SHR,UNIT=&TUNIT,
//          VOL=(,RETAIN,,,SER=&TVOLSER),LABEL=(6,SL)
//SYSUT2 DD DSN=&CUSTHLQ..LINKLIB,DISP=(,CATLG),UNIT=&DUNIT,
//          SPACE=(CYL,(30,5,10)),DSNTYPE=PDS
//SYSIN DD DUMMY
/*
/* DON'T GET SYSPRT FROM TAPE - JUST ALLOCATE SPACE FOR IT
/*      FOR LATER COMPILES.
/*

```

```

//SYSPRINT EXEC PGM=IEFBR14
//SYSPRINT DD SYSOUT=*
//DD1      DD DSN=&CUSTHLQ..SYSPRINT,DISP=(,CATLG),UNIT=&DUNIT,
//          SPACE=(CYL,(25,25,100)),DSNTYPE=&PDSTYPE
//SYSIN     DD DUMMY
//*
//DBRMLIB EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=&HLQ..DBRMLIB,DISP=SHR,UNIT=&TUNIT,
//          VOL=(,RETAIN,,,SER=&TVOLSER),LABEL=(7,SL)
//SYSUT2 DD DSN=&CUSTHLQ..DBRMLIB,DISP=(,CATLG),UNIT=&DUNIT,
//          SPACE=(CYL,(1,1,5)),DSNTYPE=&PDSTYPE
//SYSIN     DD DUMMY
//*
//DEFLIB EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=&HLQ..RPEX1.DEFLIB,DISP=SHR,UNIT=&TUNIT,
//          VOL=(,RETAIN,,,SER=&TVOLSER),LABEL=(8,SL)
//SYSUT2 DD DSN=&CUSTHLQ..RPEX1.DEFLIB,DISP=(,CATLG),UNIT=&DUNIT,
//          SPACE=(CYL,(3,1,15)),DSNTYPE=&PDSTYPE
//SYSIN     DD DUMMY
//*
//DDTLIB EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=&HLQ..RPEX1.DDTLIB,DISP=SHR,UNIT=&TUNIT,
//          VOL=(,RETAIN,,,SER=&TVOLSER),LABEL=(9,SL)
//SYSUT2 DD DSN=&CUSTHLQ..RPEX1.DDTLIB,DISP=(,CATLG),UNIT=&DUNIT,
//          SPACE=(CYL,(2,2,15)),DSNTYPE=&PDSTYPE
//SYSIN     DD DUMMY
//*
//FAPLIB EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=&HLQ..RPEX1.FAPLIB,DISP=SHR,UNIT=&TUNIT,
//          VOL=(,RETAIN,,,SER=&TVOLSER),LABEL=(10,SL)
//SYSUT2 DD DSN=&CUSTHLQ..RPEX1.FAPLIB,DISP=(,CATLG),UNIT=&DUNIT,
//          SPACE=(CYL,(2,2,15)),DSNTYPE=&PDSTYPE
//SYSIN     DD DUMMY
//*
//COMPLIB EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=&HLQ..RPEX1.COMPLIB,DISP=SHR,UNIT=&TUNIT,
//          VOL=(,RETAIN,,,SER=&TVOLSER),LABEL=(11,SL)
//SYSUT2 DD
DSN=&CUSTHLQ..RPEX1.COMPLIB,DISP=(,CATLG),UNIT=&DUNIT,
//          SPACE=(CYL,(2,2,15)),DSNTYPE=&PDSTYPE
//SYSIN     DD DUMMY
//*
//*      NOTE THAT IEBCOPY IS USED TO COPY THE EXTRACT FILE, WHICH
//*      IS A SEQUENTIAL FILE, NOT A PDS.
//*
//EXTRACT EXEC PGM=IEBGENER
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=&HLQ..RPEX1.EXTRACT,DISP=SHR,UNIT=&TUNIT,
//          VOL=(,RETAIN,,,SER=&TVOLSER),LABEL=(12,SL)
//SYSUT2 DD
DSN=&CUSTHLQ..RPEX1.EXTRACT,DISP=(,CATLG),UNIT=&DUNIT,

```

```

//          SPACE=(CYL,(1,1))
//SYSIN     DD DUMMY
//*
//LOG2IMG EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=&HLQ..RPEX1.LOG2IMG,DISP=SHR,UNIT=&TUNIT,
//          VOL=(,RETAIN,,,SER=&TVOLSER),LABEL=(13,SL)
//SYSUT2 DD
DSN=&CUSTHLQ..RPEX1.LOG2IMG,DISP=(,CATLG),UNIT=&DUNIT,
//          SPACE=(CYL,(1,1,5)),DSNTYPE=&PDSTYPE
//SYSIN     DD DUMMY
//*
//LOG2XFNT EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=&HLQ..RPEX1.LOG2XFNT,DISP=SHR,UNIT=&TUNIT,
//          VOL=(,RETAIN,,,SER=&TVOLSER),LABEL=(14,SL)
//SYSUT2 DD
DSN=&CUSTHLQ..RPEX1.LOG2XFNT,DISP=(,CATLG),UNIT=&DUNIT,
//          SPACE=(CYL,(1,1,5)),DSNTYPE=&PDSTYPE
//SYSIN     DD DUMMY
//*
//PFRMLIB EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=&HLQ..RPEX1.PFRMLIB,DISP=SHR,UNIT=&TUNIT,
//          VOL=(,RETAIN,,,SER=&TVOLSER),LABEL=(15,SL)
//SYSUT2 DD
DSN=&CUSTHLQ..RPEX1.PFRMLIB,DISP=(,CATLG),UNIT=&DUNIT,
//          SPACE=(CYL,(1,1,5)),DSNTYPE=&PDSTYPE
//SYSIN     DD DUMMY
//*
//DEFLIBU EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=&HLQ..UTEX1.DEFLIB,DISP=SHR,UNIT=&TUNIT,
//          VOL=(,RETAIN,,,SER=&TVOLSER),LABEL=(16,SL)
//SYSUT2 DD DSN=&CUSTHLQ..UTEX1.DEFLIB,DISP=(,CATLG),UNIT=&DUNIT,
//          SPACE=(CYL,(3,1,15)),DSNTYPE=&PDSTYPE
//SYSIN     DD DUMMY
//*
//DDTLIBU EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=&HLQ..UTEX1.DDTLIB,DISP=SHR,UNIT=&TUNIT,
//          VOL=(,RETAIN,,,SER=&TVOLSER),LABEL=(17,SL)
//SYSUT2 DD DSN=&CUSTHLQ..UTEX1.DDTLIB,DISP=(,CATLG),UNIT=&DUNIT,
//          SPACE=(CYL,(2,2,15)),DSNTYPE=&PDSTYPE
//SYSIN     DD DUMMY
//*
//FAPLIBU EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=&HLQ..UTEX1.FAPLIB,DISP=SHR,UNIT=&TUNIT,
//          VOL=(,RETAIN,,,SER=&TVOLSER),LABEL=(18,SL)
//SYSUT2 DD DSN=&CUSTHLQ..UTEX1.FAPLIB,DISP=(,CATLG),UNIT=&DUNIT,
//          SPACE=(CYL,(2,2,15)),DSNTYPE=&PDSTYPE
//SYSIN     DD DUMMY
//*
//COMPLIBU EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*

```

```

//SYSUT1 DD DSN=&HLQ..UTEX1.COMPLIB,DISP=SHR,UNIT=&TUNIT,
//          VOL=(,RETAIN,,SER=&TVOLSER),LABEL=(19,SL)
//SYSUT2 DD
DSN=&CUSTHLQ..UTEX1.COMPLIB,DISP=(,CATLG),UNIT=&DUNIT,
//          SPACE=(CYL,(2,2,15)),DSNTYPE=&PDSTYPE
//SYSIN    DD DUMMY
//*
//*      NOTE THAT IEBCGENER IS USED TO COPY THE EXTRACT FILE, WHICH
//*      IS A SEQUENTIAL FILE, NOT A PDS.
//*
//EXTRACTU EXEC PGM=IEBCGENER
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=&HLQ..UTEX1.EXTRACT,DISP=SHR,UNIT=&TUNIT,
//          VOL=(,RETAIN,,SER=&TVOLSER),LABEL=(20,SL)
//SYSUT2 DD
DSN=&CUSTHLQ..UTEX1.EXTRACT,DISP=(,CATLG),UNIT=&DUNIT,
//          SPACE=(CYL,(1,1))
//SYSIN    DD DUMMY
//*
//LOG2IMGU EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=&HLQ..UTEX1.LOG2IMG,DISP=SHR,UNIT=&TUNIT,
//          VOL=(,RETAIN,,SER=&TVOLSER),LABEL=(21,SL)
//SYSUT2 DD
DSN=&CUSTHLQ..UTEX1.LOG2IMG,DISP=(,CATLG),UNIT=&DUNIT,
//          SPACE=(CYL,(1,1,5)),DSNTYPE=&PDSTYPE
//SYSIN    DD DUMMY
//*
//LOG2XFTU EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=&HLQ..UTEX1.LOG2XFNT,DISP=SHR,UNIT=&TUNIT,
//          VOL=(,RETAIN,,SER=&TVOLSER),LABEL=(22,SL)
//SYSUT2 DD
DSN=&CUSTHLQ..UTEX1.LOG2XFNT,DISP=(,CATLG),UNIT=&DUNIT,
//          SPACE=(CYL,(1,1,5)),DSNTYPE=&PDSTYPE
//SYSIN    DD DUMMY
//*
//*      FORCE AFP FONTLIB TO BE PDS - PDSE HAS PROBLEMS
//*
//AFP240 EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=FSI.AGFA.AFP240.FONTLIB,DISP=SHR,UNIT=&TUNIT,
//          VOL=(,RETAIN,,SER=&TVOLSER),LABEL=(23,SL)
//SYSUT2 DD DSN=&CUSTHLQ..AGFA.AFP240.FONTLIB,DISP=(,CATLG),
//          UNIT=&DUNIT,
//          SPACE=(CYL,(35,10,150)),DSNTYPE=PDS
//SYSIN    DD DUMMY
//*
//*      FORCE AFP FONTLIB TO BE PDS - PDSE HAS PROBLEMS
//*
//AFP300 EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=FSI.AGFA.AFP300.FONTLIB,DISP=SHR,UNIT=&TUNIT,
//          VOL=(,RETAIN,,SER=&TVOLSER),LABEL=(24,SL)
//SYSUT2 DD DSN=&CUSTHLQ..AGFA.AFP300.FONTLIB,DISP=(,CATLG),
//          UNIT=&DUNIT,

```

```
//          SPACE=(CYL,(40,10,150)),DSNTYPE=PDS
//SYSIN      DD DUMMY
//*
//XEROX EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=FSI.AGFA.XEROX.FONTLIB,DISP=SHR,UNIT=&TUNIT,
//          VOL=(,RETAIN,,SER=&TVOLSER),LABEL=(25,SL)
//SYSUT2 DD DSN=&CUSTHLQ..AGFA.XEROX.FONTLIB,DISP=(,CATLG),
//          UNIT=&DUNIT,
//          SPACE=(CYL,(25,5,75)),DSNTYPE=&PDSTYPE
//SYSIN      DD DUMMY
//XEROXR EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=FSI.AGFA.XEROX.ROTATED.FONTLIB,DISP=SHR,UNIT=&TUNIT,
//          VOL=(,RETAIN,,SER=&TVOLSER),LABEL=(26,SL)
//SYSUT2 DD DSN=&CUSTHLQ..AGFA.XEROX.ROTATED.FONTLIB,DISP=(,CATLG),
//          UNIT=&DUNIT,
//          SPACE=(CYL,(115,10,500)),DSNTYPE=&PDSTYPE
//SYSIN      DD DUMMY
//*
//          PEND
// * * * * *
// *
//STEP1 EXEC LOAD,TUNIT=CART,
//          TVOLSER=D10301,          <= CHANGE
//          CUSTHLQ='FSI.v103',      <= CHANGE
//          DUNIT=SYSDA,
//          PDSTYPE=LIBRARY
//*
```

OS/390 LIBRARIES

The system begins as many source code modules which are separated into directories or libraries. Each library performs a specific task. For instance, INILIB handles Initialization file tasks and GVMLIB performs global variable manager functions.

Some of these libraries perform functions that are relevant only on the PC platform. You do not need to upload these libraries to your MVS system. The libraries you must upload to your MVS system are listed in the table below, along with the various print drivers.

An asterisk (*) indicates the library is needed only to perform a particular function. For instance FAP2MET is required only if you are converting FAPs to PMETs and running with the CompileInstream INI option set *to* **o**.

Library	Required for the system?	Required for Xerox?	Required for AFP?	Required for Postscript?	Required for PCL?
AFP2MVS			Y		
AFPCOPY			Y		
AFPFMDEF			Y		
AFPLIB			Y		
AFPPRT			Y		
ARCLIB	Y*				
ASCLIB	Y				
BARR2MVS					
BARRWRAP		Y*			
CARLIB	Y				
CFA2FAP	Y				
CPCNV	Y				
CUSLIB	Y				
CUSLIB	Y				
DALLIB	Y				
DB2LIB	Y*				
DBLIB	Y				
DSLIB	Y				
ERRLIB	Y				
FAP2CFA					
FAP2MET		Y*			

Library	Required for the system?	Required for Xerox?	Required for AFP?	Required for Postscript?	Required for PCL?
FAP2OVL			Y*		
FAPLIB	Y				
FDT2CFA	Y				
FIXOFFS					
FNTLIB	Y				
FSILIB	Y				
FSIVER	Y				
GenArc	Y*				
GenData	Y				
GENLIB	Y				
GenPrint	Y				
GenTrn	Y				
GLBLIB	Y				
GRFLIB	Y				
GVMLIB	Y				
INILIB	Y				
LBYLIB	Y				
LOG2PSEG			Y*		
LOGLIB	Y				
METOPT		Y			
PCLLIB					Y
PDS2VSAM	Y*				
PRTLIB	Y				
PSTLIB				Y	
RCBLIB	Y				
RCP2VSAM	Y*				
RPLIB	Y				

Library	Required for the system?	Required for Xerox?	Required for AFP?	Required for Postscript?	Required for PCL?
RULLIB	Y				
SAMPAFP			Y		
TIFLIB	Y				
TRANSLAT	Y				
UTILLIB	Y				
UTLLIB	Y				
VMMLIB	Y				
XDBLIB	Y				
XERDNLD		Y*			
XERLIB		Y			
XMCLIB		Y			

UPLOADING AND DOWNLOADING RESOURCES

The standard directory structure for resources on the PC is:

```
C:\RPEX1\
  DEFLIB\
  DDTLIB\
  COMPLIB\
  FORMS\
  EXTRACT\
```

These resources exist on the PC as text files and are sometimes referred to as *ASCII files* (COMPLIB is an exception and will be explained shortly). The resources are generally uploaded into files on an MVS as follows:

PC Directory	MVS File Name	MVS File Type
DEFLIB*.*	&HLQ..&RES..DEFLIB(*)	PDS
DDTLIB*.*	&HLQ..&RES..DDTLIB(*)	PDS
FORMS*.*	&HLQ..&RES..FAPLIB(*)	PDS
EXTRACT*.*	&HLQ..&RES..EXTRACT	Sequential file

Text files on a PC are represented using the ASCII character set. Text files on an MVS are represented using the EBCDIC character set. As these text files are uploaded from the PC to an MVS, each text character must be translated from ASCII to EBCDIC. The program that provides the communications between your PC and MVS (3270 Emulator) generally includes a file transfer feature. Included in the file transfer feature is an option to translate files from ASCII to EBCDIC as the files are uploaded.

COMPLIB stands for *Compiled Resources Library*. FAP files can be parsed, *compiled*, before you run the system, and placed into the COMPLIB. The utilities you can use to compile FAP files are: FAP2CFA and FDT2CFA.

This compilation improves performance since the text-oriented FAP files do not have to be parsed again during the print assembly process.

NOTE You *must*, however, run the utility using the same version and on the same platform on which you will run the system—you *cannot* compile the FAP files on the PC and upload the resulting CFA files onto an MVS system. You *must* run the FAP2CFA or FDT2CFA utilities on your MVS system.

TRANSFERRING FILES

Your 3270 emulation program may include features which let you transfer files from the PC to the MVS (upload) and from the MVS to the PC (download). Part of this capability will probably be an option to transfer *binary* file or *atext* file. Generally, when a 3270 emulator uploads a binary file, it does not translate characters as the file is uploaded—it uploads the file as is. What's more, any carriage return and line feed (CRLF) characters are also uploaded as is.

Conversely, when a 3270 emulation program uploads a file designated as text, it translates each character as it uploads it and removes any CRLF characters from the file. Characters *between* the CRLFs are assumed to be the records and are placed into the file on MVS. You will almost always upload your resources to the MVS *as text* files (see the following topic, [Handling International Characters on page 74](#) for an exception).

Your 3270 emulation program may also let you upload a file from the PC to MVS using the DOS Command Line interface. The upload program is usually called SEND and the download program is called RECEIVE.

For example, to upload the PC file “myfap fap” in the c:\rpex1\forms directory to the MVS PDS “DOC.FAPLIB” you could enter the following command in a DOS window:

SEND c:\rpex1\forms\myfap.fap 'doc.faplib(myfap)' ASCII CRLF

This SEND command include the exact file name and full path name of the PC file followed by a space and then the name of the host file ('doc.faplib(myfap)'). In this case the host file is a PDS (Partioned Data Set) so you must indicate what member to place the file into.

The *ASCII* parameter tells the system to translate the PC file from ASCII to EBCDIC as it's uploaded. The *CRLF* parameter tells the system to remove the carriage return and line feed characters as it's uploaded.

Using the ASCII and CRLF parameters tells the SEND command to treat the file as a file. If you omitted the ASCII and CRLF parameters, the SEND command would treat the file as *abinary* file. There are some situations where you might want to use just the ASCII parameter or just the CRLF parameter, but these situations are rare ([Handling International Characters on page 74](#) for an example) and you will generally use either both of them or neither of them.

HANDLING INTERNATIONAL CHARACTERS

Docucorp uses the Windows ANSI code page (1004) for files residing on the PC and the EBCDIC code page 37 for files residing on an MVS system. There are, however, characters in code page 1004 which are not in code page 37—mainly desktop publishing characters from code point 128 to 159.

To support these characters, the system uses undefined code points in code page 37 (code points below 64). For maximum portability, avoid using characters which are not defined in code page 37.

If you have resource files, such as extract or FAP files, on the PC which contain international characters you will need to translate those characters into the correct code point when you upload them from the PC to an MVS system.

Depending on your 3270 emulation program, you might be able to tell it to translate the file during the upload. If your 3270 emulation program does not let you specify the code page to translate from and to during the upload, or if your resource files on the PC use some of the undefined code points in code page 37, use the CPCNV utility.

The CPCNV utility converts text files from one code page to another. You can run the CPCNV utility either on the PC or on an MVS system. Below are examples of how you would convert a FAP file named “french1.fap”, on a PC or on an MVS system:

Running CPCNV on a PC Follow these steps to run the CPCNV utility on a personal computer:

- 1 Run the CPCNV program on the FAP files as follows:

```
cpcnv /i=french1 /s=1004 /d=37
```

- 2 Upload the FAP files as follows:

```
SEND french1.fap 'doc.faplib(french1)' CRLF
```

Note that this is one of the cases in which you use the CRLF parameter but do not use the ASCII parameter. You do not use the ASCII parameter because you have already translated the text characters in french1.fap from ANSI code page 1004 (ASCII) to EBCDIC code page 37. You still use CRLF however, because you want the upload process to remove the carriage return and line feed characters from the file as it is uploaded.

Running CPCNV on an MVS Follow these steps to run the CPCNV utility on an MVS:

- 1 Upload the FAP as follows:

```
SEND french1.fap 'doc.faplib.unconv(french1)' CRLF
```

Again, note that you do not use ASCII, but you do use CRLF. In this case, you do not want the 3270 Emulator program to translate from ASCII to EBCDIC because you are going to let the CPCNV utility on an MVS system perform the translation. Also, note that the file name on the MVS system ‘doc.faplib.unconv’ denotes that the resources in this PD are not converted.

- 2 Run the CPCNV utility on the FAP file using, as an example, the JCL provided in member CPCNVX of JCLLIB. Just as on the PC, the source code page will be 1004 and the destination code page will be 37.

XEROX IMAGE, FONT, AND FORM FILES

You can use the XERDNLD utility to send a resource to the Xerox printer and save it on the printer's disk drive. Some of the Xerox resources you might want to save to the printer are Xerox images (IMG), fonts (FNT) and forms (FRM). Each of these resources must first be created on the PC and then uploaded to the MVS system ~~without~~ ^{without} ASCII or CRLF.

To get a Xerox resource from the PC up to the MVS system and then to a Xerox printer, follow these steps:

- 1 Define a PDS on the MVS system to upload into. Define the PDS with the DCB characteristics as shown in the XERLOGOS job in JCLLIB, such as RECFM=VBM, LRECL=512, BLKSIZE=23000.
- 2 Upload the Xerox resource from the PC to the MVS system ~~without~~ ^{without} ASCII or CRLF.
- 3 Run the XERDNLD utility against this resource. There is a sample job in JCLLIB called XERDNLDX which runs XERDNLD. Note the output PDS is again defined with the characteristics RECFM=VBM, LRECL=512, and BLKSIZE=23000.
- 4 Run the GENERXER job in JCLLIB. Change the SYSUT1 DD statement so it points to the output from the XERDNLD utility, as discussed in step 3.

Change the SYSUT2 DD statement's SYSOUT class to the correct class for the Xerox printer. The GENERXER job will copy the resource into this SYSOUT class and the resource will be saved onto the Xerox printer's disk drive. Before you send the resource to the printer, start the Xerox printer in the same manner you currently start it (or intend to start) to print Documaker Server output.

XEROX PRECOMPILED METACODE (PMET) FILES

Currently, you cannot upload Xerox pre-compiled Metacode (PMET) files generated on the PC onto an MVS system. You must run the FAP2MET utility on an MVS system and convert the FAPs to PMETs on your MVS system.

OS/390 PROGRAMS AND UTILITIES

The following is a list and a short description of the programs and utilities you can run on an OS/390 (MVS) system. For more information about these utilities, see the [Docutoolbox Reference](#)

- AFPCOPY utility** This utility takes an AFP print stream which may contain multiple BEGINDOC and ENDDOC statements and generates an AFP print stream with only one BEGINDOC and one ENDDOC statement—in other words, it converts the print stream ~~from~~*from sheet to continuous-form*. No parameters are required. The input AFP print stream is read from the INFILE DD statement and the output AFP print stream is written to the OUTFILE DD statement. Look in member AFPCOPYX of JCLLIB to find an example of this utility.
- AFPFMDEF utility** This utility accepts as input a source form definition file and generates an AFP FormDef. Look in member AFPFMDFX of JCLLIB to find an example of this utility.
- AFP2MVS utility** This utility accepts as input an AFP resource, such as a font, page segment, or print-ready file, that's been uploaded to the MVS system from the PC with no ASCII and no CRLF. It also generates that resource to another file such that the record-oriented format of the resource will be restored. Look in the AFP2MVSX member of JCLLIB to find an example of this utility.
- BARR2MVS utility** This utility program accepts as input a print-ready Metacode file produced by Documaker Server running on the PC (using these INI options: Environment=OS22, OutMode=BARR, or BARRWORD), strips that file of the BARR-encoded information and produces a Metacode print-ready file acceptable in an MVS environment. Look in the BARR2MVX member of JCLLIB to find an example of this utility.
- BARRWRAP utility** This utility accepts as input a print-ready Metacode file produced by Documaker Server on MVS and produces a Metacode print-ready ~~file~~*file wrapped* with control information the BARR printer-attachment system understands. Look in the BARRWRPX member of JCLLIB to find an example of this utility.
- CPCNV utility** This utility converts a source file from one code page to another. The utility reads an input source file, such as an extract file, a FAP file, and so on, which corresponds to a specific code page. You can convert either a single member in the PDS (use /I=member) or all members of the PDS (use /I=*). The utility then generates a target file which corresponds to a different code page. Look in member CPCNVX of JCLLIB to find an example of this utility.
- FAP2CFA utility** This utility accepts as input a FAP file and generates a compiled FAP file. Look in member FAP2CFAX of JCLLIB to find an example of this utility.
- FAP2MET utility** This utility accepts as input a FAP file and generates a pre-compiled Metacode member, usually placing the ~~PME~~*PME* into the PMETLIB dataset. You can convert a single FAP in the PDS (use /I=fapname) or all FAPs in the PDS (use /I=*). There are several options you can select when you run the FAP2MET utility. For example, you can save the PMET in the PMETLIB dataset so it is ready to be included by the GenPrint program into a print stream or you can generate the PMET so it can be sent directly to the Metacode printer. You can also select the HMI, Short Edge Bind, or Long Edge Bind options. Look in member FAP2METX of JCLLIB to find an example of this utility.